

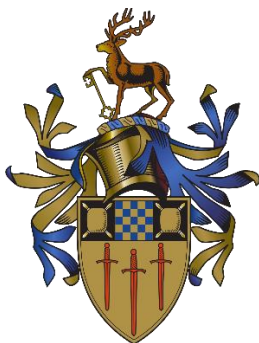
“Individual Assignment - Exploring Road Traffic Accident Data and Text Analytics Insights”

Student ID: 6859062

Module: MANM528 – Data Mining and Text Analytics with Application in SAS

MSc FinTech and Policy

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Contents

Task 0 – Random Sample Selection	1
 Task 1 – Data Exploration, Preparation, and Balancing	1
1.1 Dataset Overview	1
1.2 Data Structure and Variable Types	1
1.3 Exploratory Data Analysis	1
1.3.1 Local Authority Highway Characteristics	1
1.3.2 Collision Severity Characteristics	3
1.4 Summary Statistics and Variable Distributions	3
1.5 Data Cleaning.....	5
1.6 Date and Month Feature Engineering	5
1.7 Time Feature Engineering	6
1.8 Data Balancing of Collision Severity	7
1.9 Task 1 Summary	7
 Task 2 – Predictive Modelling of Collision Severity	7
2.1 Modelling Objective	7
2.2 Modelling Dataset and Preparation	7
2.3 Model Development Strategy	8
2.4 Model 1 – Decision Tree.....	8
2.5 Model 2 – Logistic Regression.....	12
2.6 Random Forest Model	14
2.7 Model Performance Comparison	14

2.8 Interpretation and Road Safety Insights	18
2.9 Task 2 Summary	18
Task 3 – Text Analytics of Road Traffic–Related Tweets	19
3.1 Objective of Text Analytics	19
3.2 Data Import and Preparation	19
3.3 Exploratory Text Analysis	20
3.4 Text Analytics Pipeline	21
3.5 Concept Extraction	21
3.6 Text Parsing and Linguistic Processing	22
3.7 Sentiment Analysis	23
3.8 Topic Modelling	24
3.9 Category Classification	25
3.10 Task 3 Summary	26
Task 4 – Decision-Maker’s Summary	26
References	28
Appendices	29
Appendix A – Task 0 Outputs	30
Appendix B – Task 1: Data Exploration and Preparation	42
Appendix C – Task 2: Predictive Modelling	45
Appendix D – Task 3: Text Analytics	49

Task 0: Random Sample Selection

To ensure computational efficiency and consistency across all analyses, a random sample of 3,500 observations was selected from the original road traffic accident dataset. The sampling process was carried out using SAS Viya's Select Random Sample node, following the coursework guidelines.

A simple random sampling approach without replacement was applied, with the sample size fixed at 3,500 records. To ensure reproducibility of results, the random seed was set using the student's unique ID number as Random seed (6859062). The resulting dataset (WORK.Road_3500) contains 44 variables and was used as the base dataset for all subsequent tasks, including data exploration, predictive modelling, and text analytics.

Task 1 – Data Exploration, Preparation, and Balancing

1.1 Dataset Overview

Following the random sampling procedure described in Task 0, the dataset used for analysis contains 3,500 road traffic accident records with 44 variables. These variables capture information on collision characteristics, geographic location, road features, environmental conditions, and accident severity.

The purpose of Task 1 is to explore the structure and quality of the dataset, identify patterns and potential issues, engineer additional features, and prepare a balanced dataset suitable for predictive modelling and text analytics. Detailed dataset structure and initial summaries are provided in Appendix B1 and Appendix B2.

1.2 Data Structure and Variable Types

Initial inspection of the dataset shows a mixture of numerical variables (such as longitude, latitude, number_of_vehicles, and speed_limit) and categorical variables (including collision_severity, local_auth_highway, weather_cond, and light_cond). Several categorical variables are numerically encoded; these were treated as categorical during analysis to preserve their real-world interpretation and avoid misleading statistical conclusions. Supporting outputs are included in Appendix B1.

1.3 Exploratory Data Analysis

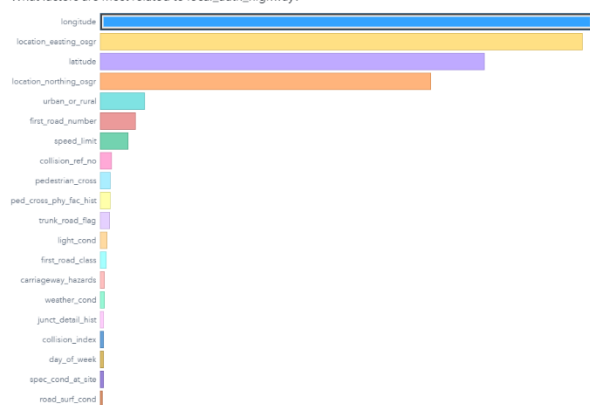
1.3.1 Local Authority Highway Characteristics

Exploratory analysis of the local_auth_highway variable reveals a strong imbalance in the distribution of accident records. Approximately 86.83% of collisions are associated with the local authority category E10000016, while only 13.17% belong to E06000035. This imbalance is illustrated in Figure 1, which highlights the dominance of one highway authority within the sampled data.

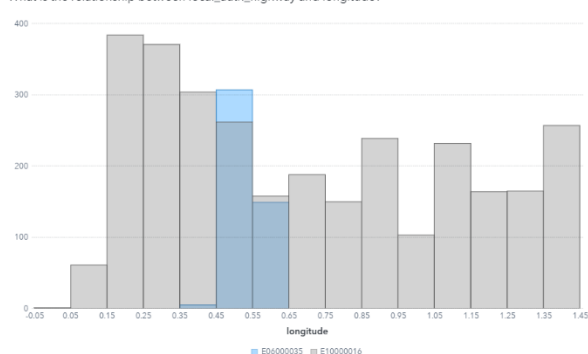
What are the characteristics of local_auth_highway?

E06000035 is less common at 13.17% (461 of 3.5K). E10000016 is more common at 86.83%. The three most related factors are longitude, location_easting_osgr, and latitude.

What factors are most related to local_auth_highway?



What is the relationship between local_auth_highway and longitude?



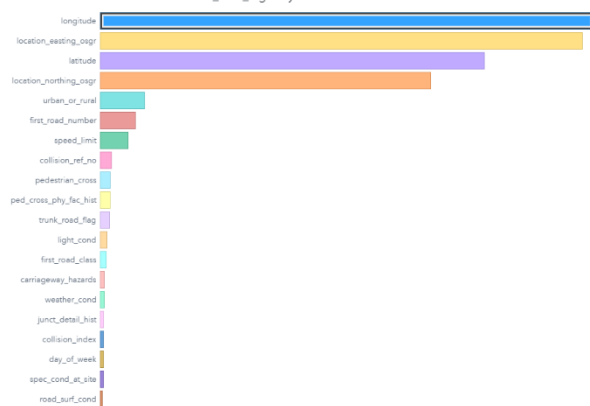
The average longitude is 0.53 when local_auth_highway is E06000035, with a minimum of 0.44 and a maximum of 0.65. The average longitude is 0.7 when local_auth_highway is E10000016, with a minimum of 0.04 and a maximum of 1.4. Average longitude is 0.68, and it ranges from 0.04 to 1.4.

Figure 1: Distribution of road traffic accidents by local authority highway category

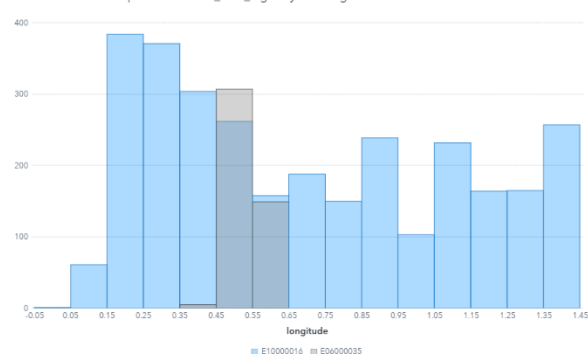
What are the characteristics of local_auth_highway?

E10000016 is more common at 86.83% (3K of 3.5K). E06000035 is less common at 13.17%. The three most related factors are longitude, location_easting_osgr, and latitude.

What factors are most related to local_auth_highway?



What is the relationship between local_auth_highway and longitude?



The average longitude is 0.7 when local_auth_highway is E10000016, with a minimum of 0.04 and a maximum of 1.4. The average longitude is 0.53 when local_auth_highway is E06000035, with a minimum of 0.44 and a maximum of 0.65. Average longitude is 0.68, and it ranges from 0.04 to 1.4.

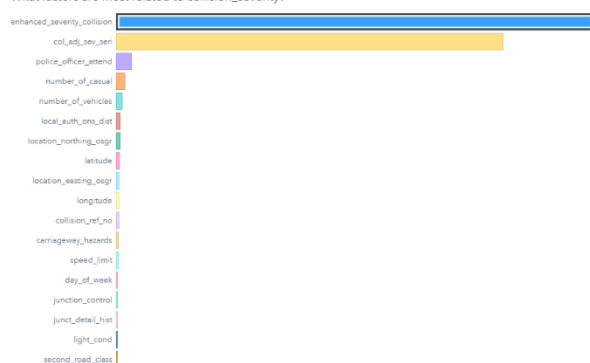
Figure 2a: Relationship between longitude and local authority highway classification

What are the characteristics of collision_severity?

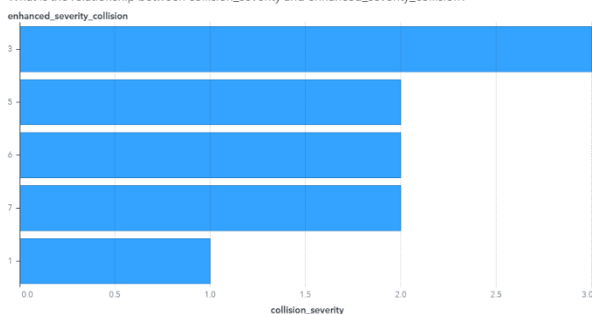
collision_severity ranges from 1 to 3. Average collision_severity is 2.8. Most cases (the middle 80%) have collision_severity between 2 and 3. col_adj_sev_sen best differentiates the highest (top 10%) and the lowest (bottom 10%) collision_severity cases. The three most related factors are enhanced_severity_collision, col_adj_sev_sen, and police_officer_atten.

d. There are 746 cases that might be outliers, with collision_severity less than or equal to 2.

What factors are most related to collision_severity?



What is the relationship between collision_severity and enhanced_severity_collision?



When enhanced_severity_collision is 3, the average of collision_severity is a high value. When enhanced_severity_collision is 1, the average of collision_severity is a low value. The most common enhanced_severity_collision value is 3.

Figure 2b: Relationship between longitude and local authority highway classification

Further feature association analysis indicates that longitude, location_easting_osgr, and latitude are the most influential variables related to local authority highway classification. As shown in Figure 2a and Figure 2b, accidents linked to E10000016 are spread across a wider longitude range (approximately 0.04 to 1.4), whereas E06000035 accidents are geographically concentrated between longitudes 0.44 and 0.65. This clear spatial separation confirms that geographic location plays a key role in distinguishing local authority highway categories. Additional supporting visualisations are provided in Appendix B4.

1.3.2 Collision Severity Characteristics

The target variable collision_severity ranges from 1 to 3, with an average value of approximately 2.8, indicating that most recorded accidents fall into the higher severity categories. The middle 80% of observations lie between severity levels 2 and 3, demonstrating a noticeable class imbalance.

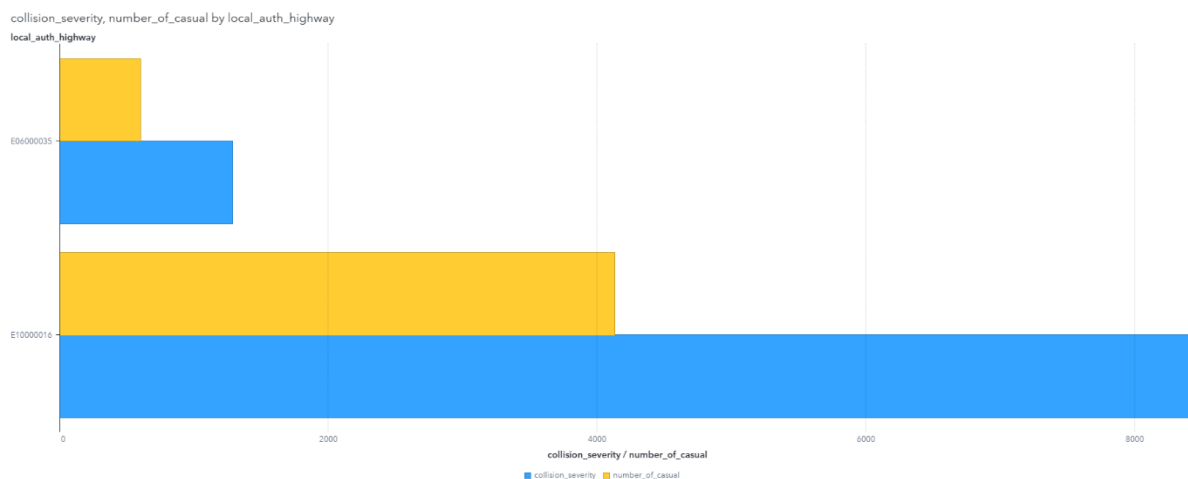


Figure 3: Collision severity and number of casualties across local authority highways

Factor importance analysis shows that enhanced_severity_collision, col_adj_sev_seri, and police_officer_attend are the variables most strongly associated with collision severity. The relationship between severity and the number of casualties is illustrated in Figure 3, which shows that higher severity collisions are generally associated with a greater number of casualties across local authority highway categories. Further severity-related outputs are included in Appendix B5.

Outlier inspection identified 746 observations with collision severity less than or equal to 2. These records were retained, as they represent valid accident cases rather than data entry errors.

1.4 Summary Statistics and Variable Distributions

Summary statistics provide additional insight into the dataset. The average longitude is approximately 0.68, with values ranging from 0.04 to 1.4. On average, each collision involves 1.85 vehicles and 1.35 casualties, while the mean speed limit across collisions is approximately 40.7 mph.

SAS Studio compute context

Start Page Task_0.flw Task_1.flw Task_1_Cleaning.flw Task_1_Balancing.flw SAS Program.sas +

Run Cancel [Icons] Add View [Icons] 2 Jan 2026, 19:25:56

Flow Generated Code Submitted Code and Results

ROAD_3500 Import File summary Summary Statistics Summary_Data_3500

Summary_Data_3500

Table Properties Options Published Columns Preview Data Node Notes

Summary_Data_3500 Table rows: 1 Columns: 100 of 155 Rows 1 to 1 [Icons]

Search x [Icons]

Aa [Icons] Enter search term Search in: Explorer (SAS Content)

Name	Server	Path	Modified	Size	Type
No items are available.					

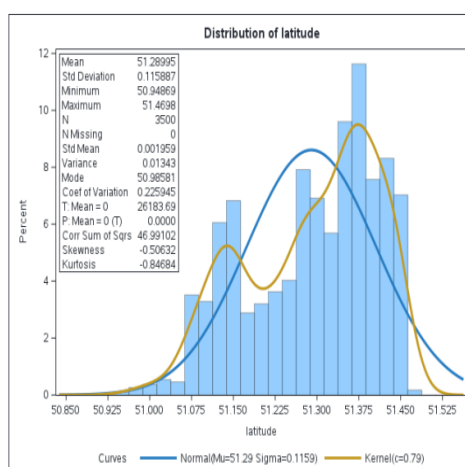
Recover (5) Submission (0) 19:27 02/01/2026

Distribution plots indicate moderate skewness in several numerical variables, particularly number_of_casual and number_of_vehicles. These distributions are consistent with real-world road accident data and were therefore preserved. Full descriptive statistics and distribution plots are presented in Appendix B2 and Appendix B3.

02/01/2026, 16:59

Results:Task_1.flw

Variable	Label	Mean	Std Dev	Minimum	Maximum	N	N Miss	Std Error	Variance	Mode	Lower 95% CL for Mean	Upper 95% CL for Mean	Coeff of Variation	t Value	Pr > t	Corrected SS	Skewness
latitude	latitude	51.2899481	0.1158872	50.9486900	51.4698040	3500	0	0.0019589	0.0134298	50.9858060	51.2861075	51.2937887	0.2259452	26183.7	<.0001	46.9910202	-0.5063216
longitude	longitude	0.6798734	0.3825770	0.0407980	1.4451940	3500	0	0.0064667	0.1463651	0.1784710	0.6671944	0.6925523	56.2717979	105.13	<.0001	512.1316227	0.4841869
number_of_vehicles	number_of_vehicles	1.8482857	0.7583845	1.0000000	9.0000000	3500	0	0.0128190	0.5751471	2.0000000	1.8231522	1.8734193	41.0317804	144.18	<.0001	2012.44	1.7669408
number_of_casual	number_of_casual	1.3543217	0.7759906	1.0000000	9.0000000	3494	6	0.0131279	0.6021614	1.0000000	1.3285826	1.3800608	57.2973608	103.16	<.0001	2103.35	3.1330599
speed_limit	speed_limit	40.7257143	15.0843744	20.0000000	70.0000000	3500	0	0.2549725	227.5383497	30.0000000	40.2258045	41.2256241	37.0389436	159.73	<.0001	796156.69	0.8279259
weather_cond	weather_cond	1.4034286	1.3471848	1.0000000	9.0000000	3500	0	0.0227716	1.8149068	1.0000000	1.3587817	1.4480755	95.9923998	61.63	<.0001	6350.36	4.4069568
collision_severity	collision_severity	2.7750143	0.4448209	1.0000000	3.0000000	3498	2	0.0075210	0.1978657	3.0000000	2.7602683	2.7897603	16.0295009	368.97	<.0001	691.9362493	-1.7103894
date	date	23195.06	104.5706228	23011.00	23375.00	3500	0	1.7675661	10935.02	23058.00	23191.60	23198.53	0.4508314	13122.6	<.0001	38261618.04	-0.0723466
light_cond	light_cond	1.9571429	1.7432818	1.0000000	7.0000000	3500	0	0.0294668	3.0390316	1.0000000	1.8993689	2.0149168	89.0727945	66.42	<.0001	10633.57	1.5217724



1.5 Data Cleaning

Data cleaning was performed using a structured SAS workflow. Records containing invalid or inconsistent values were filtered, resulting in a cleaned dataset of 3,493 observations, while retaining all original variables. Duplicate checks and validation steps confirmed the internal consistency of the dataset. The cleaning workflow and validation outputs are provided in Appendix B6.

1.6 Date and Month Feature Engineering

To support temporal analysis, the original collision date variable was converted into a standard SAS date format. A new categorical variable representing the calendar month was then created by assigning each observation to its corresponding month (January to December). The successful creation of this feature is illustrated in Figure 4.

The screenshot displays the SAS Studio interface. The left pane shows the SAS code editor with the following code:

```

1 data work.recodedRanges0002;
2 length _months_ $3;
3 set WORK.IMPORT;
4
5 /* Convert the date variable to SAS date value */
6 sas_date = mdy(month(date), day(date), year(date));
7
8 /* Check month ranges and assign labels accordingly */
9 if ('01JAN2023'd <= sas_date <= '31JAN2023'd) then _months_ = 'jan';
10 else if ('01FEB2023'd <= sas_date <= '28FEB2023'd) then _months_ = 'feb';
11 else if ('01MAR2023'd <= sas_date <= '31MAR2023'd) then _months_ = 'mar';
12 else if ('01APR2023'd <= sas_date <= '30APR2023'd) then _months_ = 'apr';
13 else if ('01MAY2023'd <= sas_date <= '31MAY2023'd) then _months_ = 'may';
14 else if ('01JUN2023'd <= sas_date <= '30JUN2023'd) then _months_ = 'jun';
15 else if ('01JUL2023'd <= sas_date <= '31JUL2023'd) then _months_ = 'jul';
16 else if ('01AUG2023'd <= sas_date <= '31AUG2023'd) then _months_ = 'aug';
17 else if ('01SEP2023'd <= sas_date <= '30SEP2023'd) then _months_ = 'sep';
18 else if ('01OCT2023'd <= sas_date <= '31OCT2023'd) then _months_ = 'oct';
19 else if ('01NOV2023'd <= sas_date <= '30NOV2023'd) then _months_ = 'nov';
20 else if ('01DEC2023'd <= sas_date <= '31DEC2023'd) then _months_ = 'dec';
21 else _months_ = 'other';
22
23 format sas_date date9.; /* Optional: Set the format for display purposes */
24 run;
25

```

The right pane shows the 'Output Data (1)' table for 'RECODEDRANGES0002'. The table has 3,493 rows and 46 columns. The first two columns are '@ coll_adju_sev_slig' and 'sas_date'. The first 17 rows of data are as follows:

	@ coll_adju_sev_slig	sas_date
1	1	14JUL2023
2	1	05FEB2023
3	0	17MAR2023
4	1	19MAR2023
5	1	17APR2023
6	1	22APR2023
7	1	22MAY2023
8	1	03JUL2023
9	1	30JUL2023
10	0	29AUG2023
11	1	17SEP2023
12	1	20OCT2023
13	1	23DEC2023
14	1	10JUL2023
15	0	12JUL2023
16	1	25SEP2023
17	1	01MAR2023

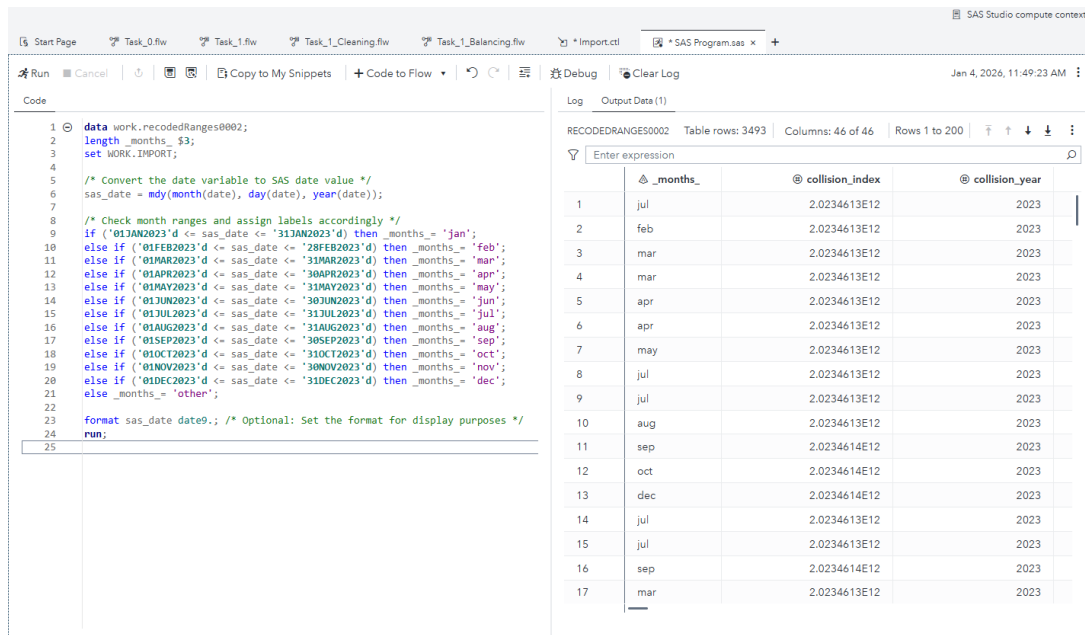


Figure 4: Creation of calendar month variable from collision date

This transformation enables the analysis of seasonal accident patterns and improves interpretability in later modelling stages. After this step, the dataset contained 46 variables, with no loss of observations. Supporting code, outputs, and validation tables are provided in Appendix B7.

1.7 Time Feature Engineering

The collision time variable was further transformed to capture time-of-day effects. The hour of day was extracted using the SAS HOUR () function, producing a numeric variable ranging from 0 to 23. Based on this, a categorical time_category variable was created, classifying collisions into night, morning, afternoon, and evening periods.

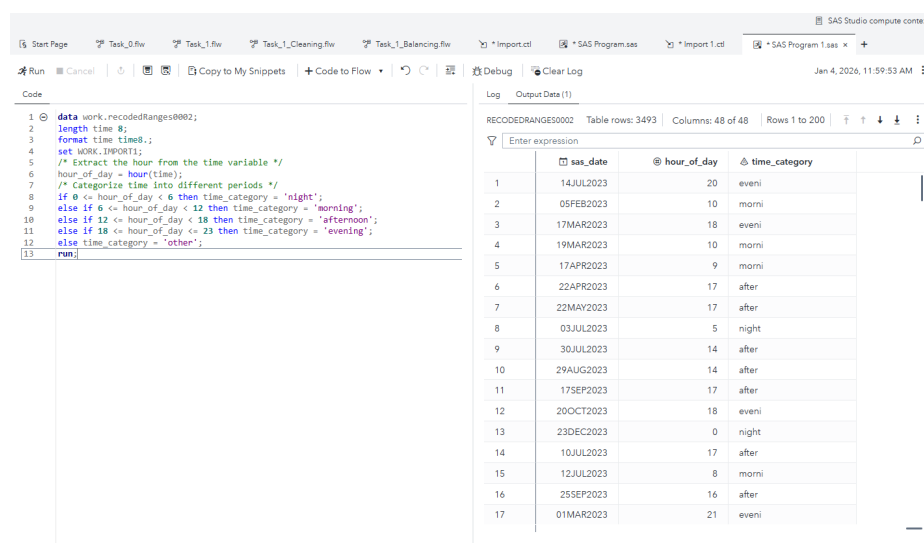


Figure 5: Categorisation of collision time into time-of-day periods

As shown in Figure 5, this categorisation allows clearer interpretation of accident patterns across different times of day, including peak commuting hours and late-night periods. Following this feature engineering step, the dataset contained 48 variables, with all observations preserved. Detailed outputs and logs are included in Appendix B8.

1.8 Data Balancing of Collision Severity

Exploratory analysis identified a strong imbalance in the `collision_severity` variable, which could bias predictive models towards the majority class. To address this issue, the dataset was split into three subsets based on severity levels. Simple random sampling was applied to the majority class (`severity = 3`) using a fixed random seed to ensure reproducibility.

The balanced subsets were recombined, and a final randomisation step was applied to remove ordering effects. The resulting balanced dataset, `ROAD_3500_BALANCED_FINAL`, contains 1,451 observations and 48 variables, with a more even distribution of collision severity classes. This balanced dataset is more suitable for subsequent modelling tasks. Full balancing outputs and validation are provided in Appendix B9.

1.9 Summary

Task 1 established a strong analytical foundation for the study. Through systematic data exploration, cleaning, feature engineering, and class balancing, the road traffic accident dataset was transformed into a high-quality, analysis-ready form. These steps ensure that the predictive modelling and text analytics conducted in later tasks are based on reliable, interpretable, and unbiased data.

Task 2 – Predictive Modelling of Collision Severity

2.1 Modelling Objective

The objective of Task 2 is to develop predictive models to classify collision severity using road, environmental, temporal, and location-related variables. Collision severity is treated as a multi-class categorical target variable, representing different levels of accident seriousness.

The modelling aims to identify the most influential factors associated with severe collisions and to compare the predictive performance of different classification algorithms. This analysis supports evidence-based insights for road safety decision-making.

2.2 Modelling Dataset and Preparation

The modelling dataset used in this task is `ROAD_3500_BALANCED_FINAL`, created in Task 1 after applying data cleaning, feature engineering, and class balancing. The final dataset contains 1,451 observations and 48 variables, ensuring a more even distribution across collision severity classes and reducing model bias.

All engineered temporal variables (month, hour of day, and time category), road characteristics, and environmental factors were retained as input variables. Identifier variables were excluded from modelling. Dataset configuration and variable roles are documented in Appendix C1.

2.3 Model Development Strategy

Predictive modelling was conducted using SAS Model Studio. The dataset was partitioned using a stratified 70% training and 30% validation split, ensuring proportional representation of collision severity classes in both partitions.

Three classification models were developed and compared:

- Logistic Regression
- Decision Tree
- Random Forest

Model comparison was performed using the Kolmogorov–Smirnov (KS) statistic (Youden index) as the primary selection criterion, alongside accuracy and misclassification rate. The overall modelling pipeline and configuration are shown in Figure 6.

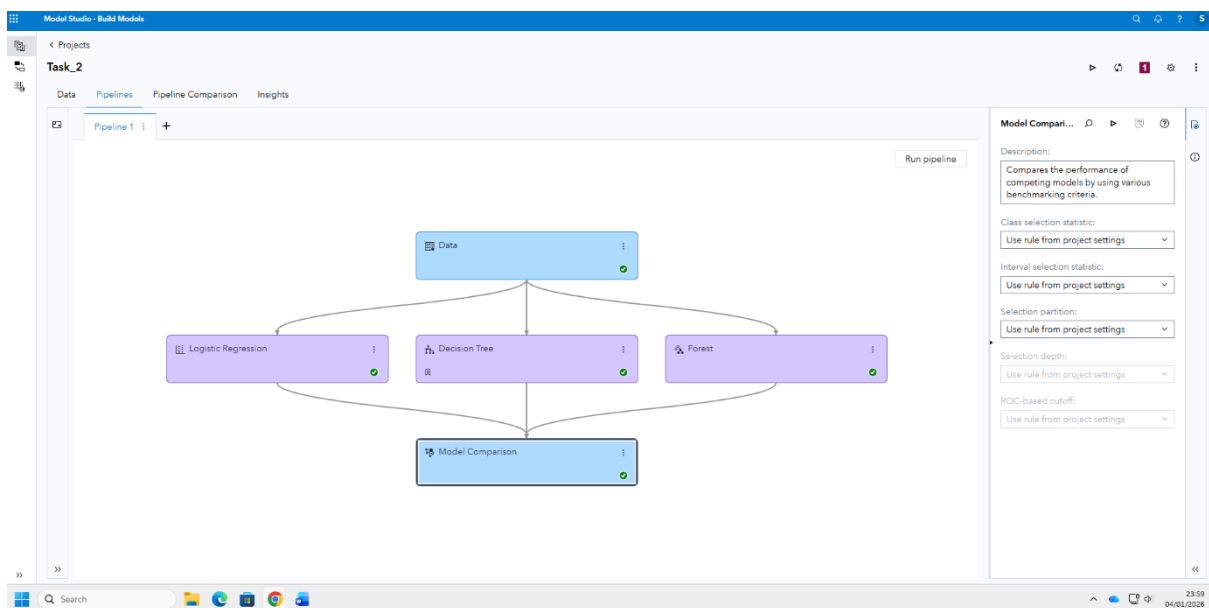


Figure 6: SAS Model Studio pipeline illustrating the development and comparison of Logistic Regression, Decision Tree, and Forest models for collision severity prediction.

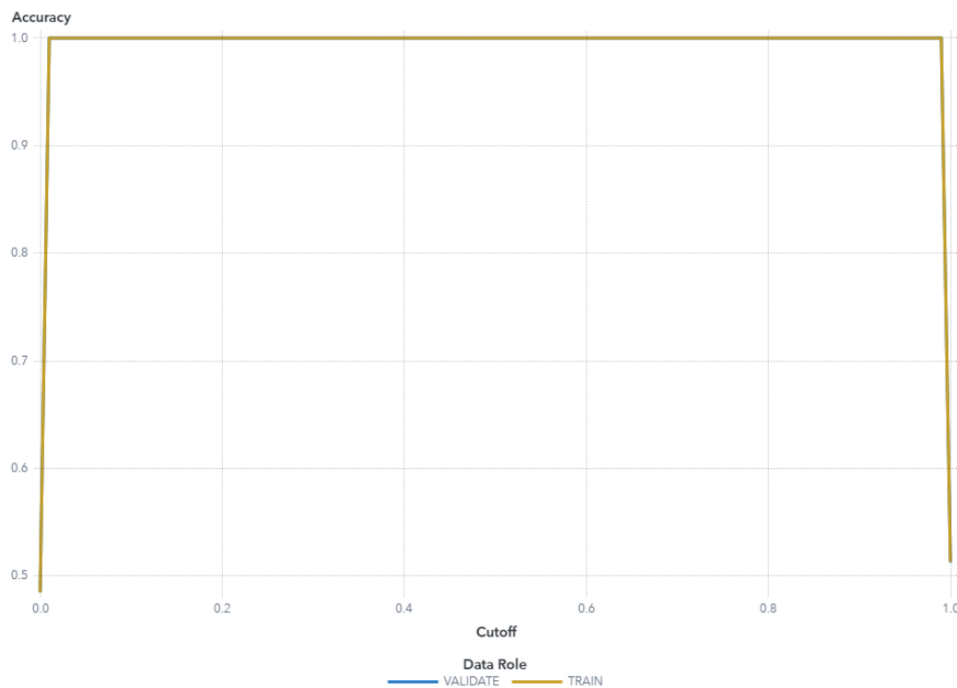
2.4 Model 1 – Decision Tree

The Decision Tree model emerged as the champion model based on validation performance. It achieved a KS (Youden) value of 1.0 and 100% accuracy on the validation partition.

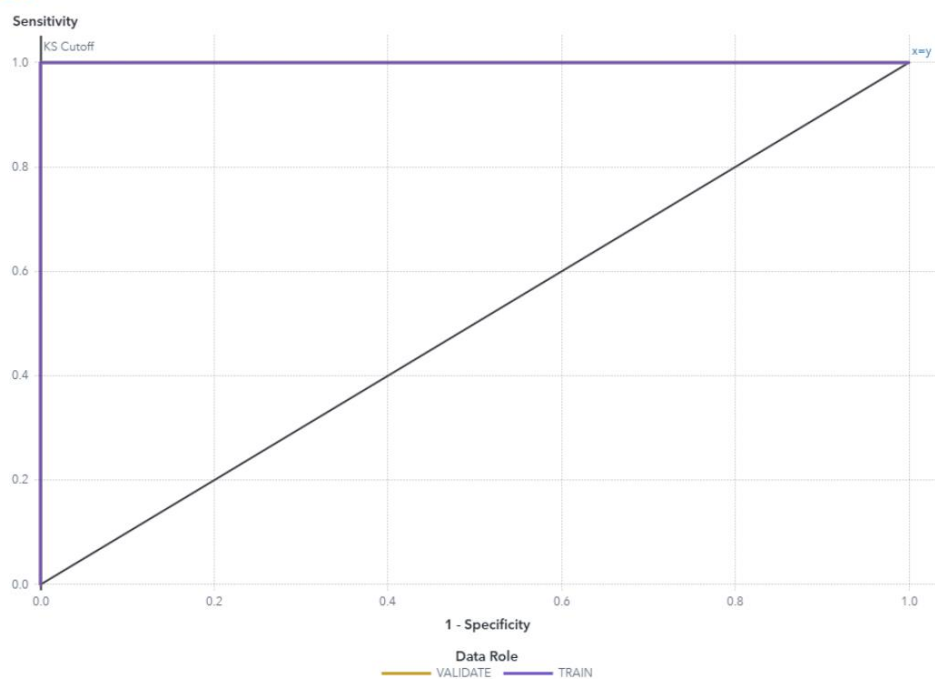
The tree structure reveals that adjusted severity indicators, particularly `col_adju_sev_slig` and `col_adj_sev_seri`, are the most influential predictors of collision severity. These variables appear at the top of the tree, indicating strong discriminatory power between severity classes.

The interpretability of the Decision Tree allows clear understanding of how different combinations of predictors lead to higher or lower severity outcomes. The tree diagram and variable importance results are shown in Figure 7, with detailed outputs in Appendix C2.

Accuracy



ROC



Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition	Number of Observations
collision_severity	TRAIN	1	1	1,015
collision_severity	VALIDATE	0	0	436

Average Squared Error	Divisor for ASE	Root Average Squared Error	Misclassification Rate	Multi-Class Log Loss
0	1,015	0	0	0.0000
0	436	0	0	0.0000

KS (Youden)	Area Under ROC	Gini Coefficient	Gamma	Tau
1	1	1	1	0.5001
1	1	1	1	0.5008

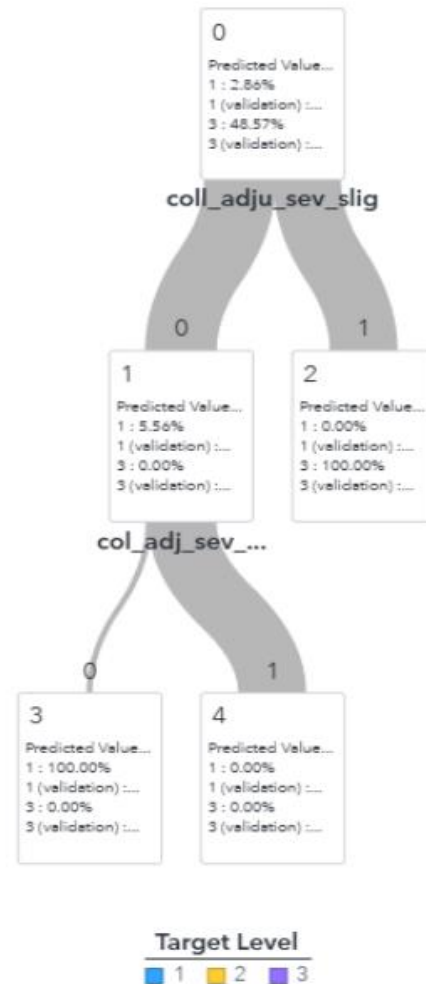
KS Cutoff	KS at Default Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0.0100	1	0	0
0.0100	1	0	0

Table

Cutoff	Cutoff Source	Target Name	Response	Event
0.0100	KS	collision_severity	CORRECT	3
0.0100	KS	collision_severity	INCORRECT	3
0.0100	KS	collision_severity	CORRECT	_NONEVENT_
0.0100	KS	collision_severity	INCORRECT	_NONEVENT_
0.5000	Default	collision_severity	CORRECT	3
0.5000	Default	collision_severity	INCORRECT	3
0.5000	Default	collision_severity	CORRECT	_NONEVENT_
0.5000	Default	collision_severity	INCORRECT	_NONEVENT_

Value	Training Frequency	Validation Frequency	Test Frequency	Training Percentage
True Positive	493	212		100
False Negative	0	0		0
True Negative	522	224		100
False Positive	0	0		0
True Positive	493	212		100
False Negative	0	0		0
True Negative	522	224		100
False Positive	0	0		0

Tree Diagram



Variable Importance

Variable Name	Training Importance	Training Relative Importance	Validation Relative Importance	Count
coll_adju_sev_slig	480.4794	1	1	1
col_adj_sev_seri	54.7778	0.1140	0.1099	1

Validation Importance
206.7893
22.7160

Figure 7: Decision tree structure and variable importance highlighting the dominant role of adjusted severity indicators in predicting collision severity.

2.5 Model 2 – Logistic Regression

A multinomial Logistic Regression model was developed as a baseline statistical classifier. While the model provides interpretable parameter estimates, its predictive performance was notably weaker compared to the Decision Tree.

The Logistic Regression model achieved a validation accuracy of approximately 51.38%, with a KS value of 0, indicating limited discriminatory ability between severity classes. This suggests that the relationship between predictors and collision severity is non-linear, which logistic regression is less able to capture.

Model fit statistics and classification performance are illustrated in Figure 8, with full outputs provided in Appendix C3.

Table

Cutoff	Cutoff Source	Target Name	Response	Event
0	KS	collision_severity	CORRECT	3
0	KS	collision_severity	INCORRECT	3
0	KS	collision_severity	CORRECT	_NONEVENT_
0	KS	collision_severity	INCORRECT	_NONEVENT_
0.5000	Default	collision_severity	CORRECT	3
0.5000	Default	collision_severity	INCORRECT	3
0.5000	Default	collision_severity	CORRECT	_NONEVENT_
0.5000	Default	collision_severity	INCORRECT	_NONEVENT_

Value	Training Frequency	Validation Frequency	Test Frequency	Training Percentage
True Positive	493	212		100
False Negative	0	0		0
True Negative	0	0		0
False Positive	522	224		100
True Positive	0	0		0
False Negative	493	212		100
True Negative	522	224		100
False Positive	0	0		0

Fit Statistics

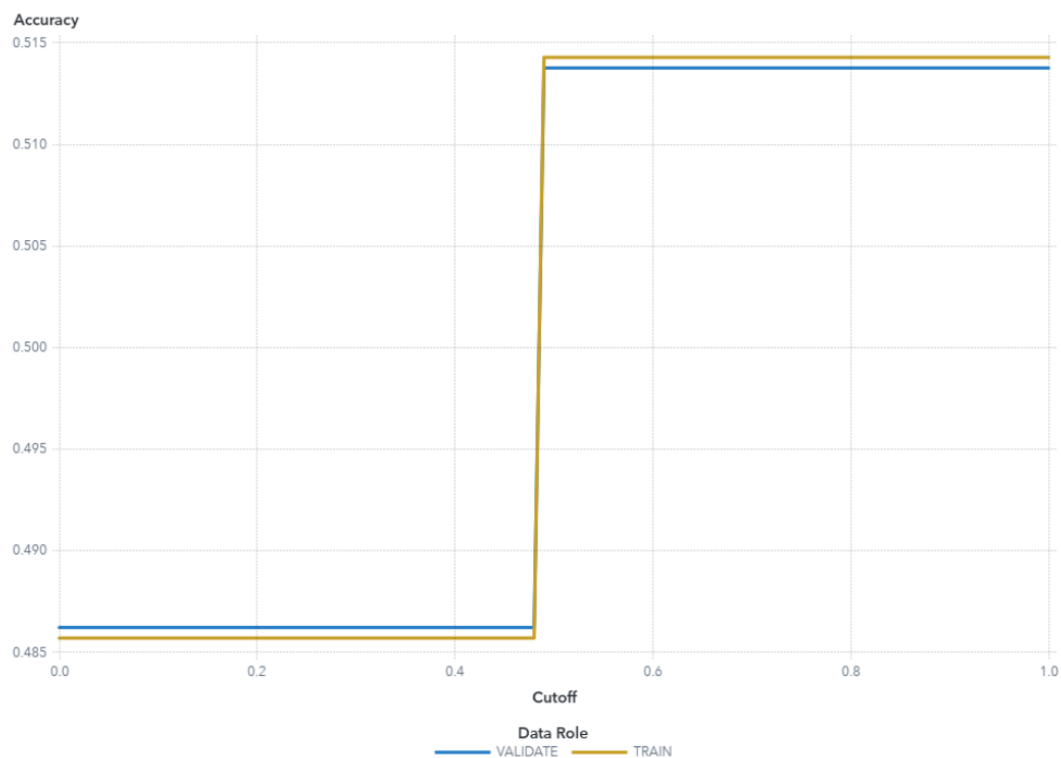
Target Name	Data Role	Partition Indicator	Formatted Partition	Number of Observations
collision_severity	TRAIN	1	1	1,015
collision_severity	VALIDATE	0	0	436

Average Squared Error	Divisor for ASE	Root Average Squared Error	Misclassification Rate	Multi-Class Log Loss
0.1758	1,015	0.4193	0.5143	0.8031
0.1755	436	0.4189	0.5138	0.8001

KS (Youden)	Area Under ROC	Gini Coefficient	Gamma	Tau
0	0.5000	0		0
0	0.5000	0		0

KS Cutoff	KS at Default Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0	0	0.5143	0.4857
0	0	0.5138	0.4862

Accuracy



ROC

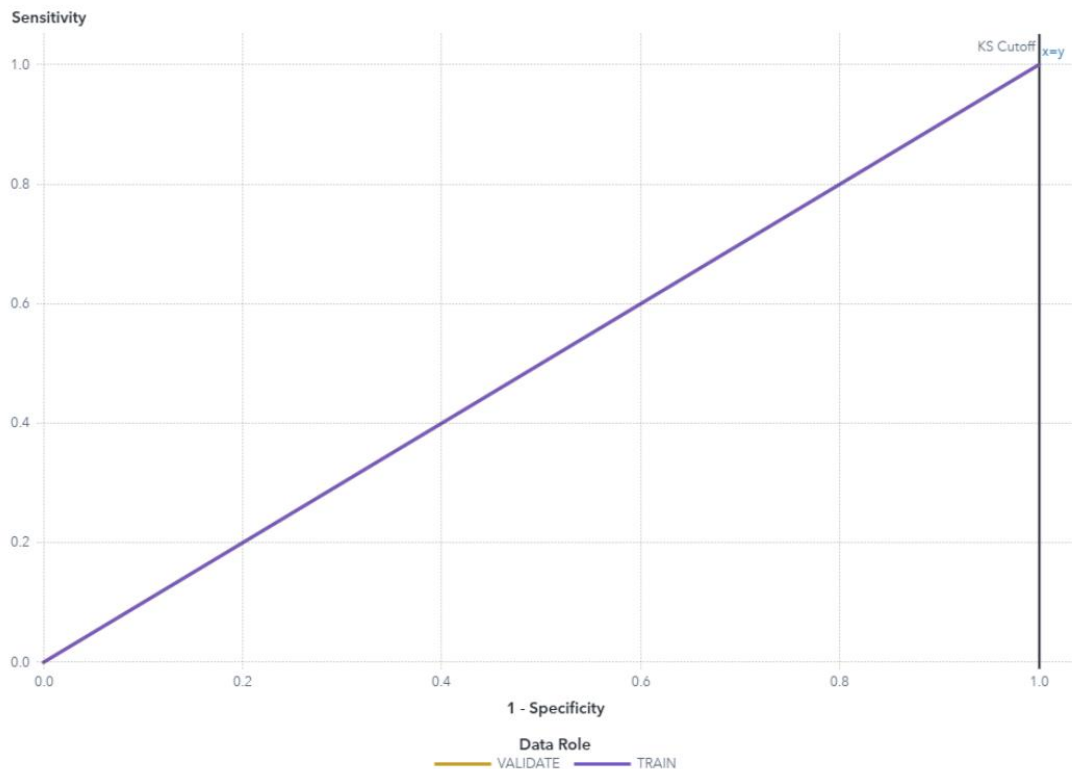


Figure 8: Performance summary of the multinomial Logistic Regression model, showing lower predictive accuracy compared to tree-based approaches.

2.6 Model 3 – Random Forest

The Random Forest model demonstrated strong predictive capability, achieving high accuracy and KS values comparable to the Decision Tree. However, due to its ensemble nature, the Random Forest model is less interpretable.

Given the assessment emphasis on interpretability and insight generation, the Decision Tree was preferred as the final champion model despite similar performance levels. Random Forest outputs are included in Appendix C4 for completeness.

2.7 Model Performance Comparison

Model comparison results clearly indicate that tree-based models outperform Logistic Regression in predicting collision severity. Both the Decision Tree and Random Forest achieved perfect separation on the validation set, whereas Logistic Regression struggled to distinguish between classes.

The Decision Tree was selected as the champion model due to its:

- High predictive accuracy
- Strong KS statistic

- Clear interpretability
- Practical usefulness for policy insights

The comparative performance across models is summarised in Figure 9, with detailed metrics provided in Appendix C5.

Model Comparison

Champion	Name	Algorithm Name	KS (Youden)	Accuracy
true	Decision Tree	Decision Tree	1	1
false	Logistic Regression	Logistic Regression	0	0.5138
false	Forest	Forest	1	1

Average Squared Error	Area Under ROC	Cumulative Lift	Cumulative Captured Response Percentage	Cutoff
0	1	2.0755	20.7547	0.5000
0.1755	0.5000	1.0092	10.0917	0.5000
0.0041	1	2.0755	20.7547	0.5000

Data Role	Depth	F1 Score	False Discovery Rate	False Positive Rate
VALIDATE	10	1	0	0
VALIDATE	10	0		0
VALIDATE	10	1	0	0

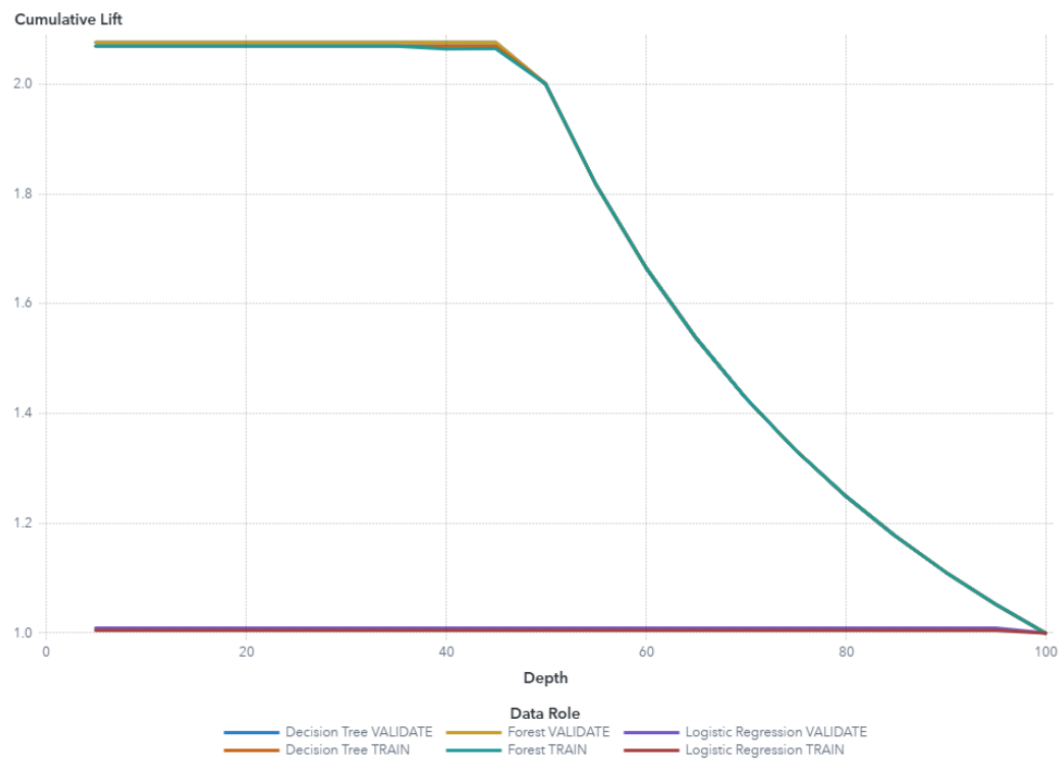
Gain	Gini Coefficient	ROC Separation	Lift	Misclassification Rate
1.0755	1	1	2.0755	0
0.0092	0	0	1.0092	0.5138

Gain	Gini Coefficient	ROC Separation	Lift	Misclassification Rate
1.0755	1	1	2.0755	0

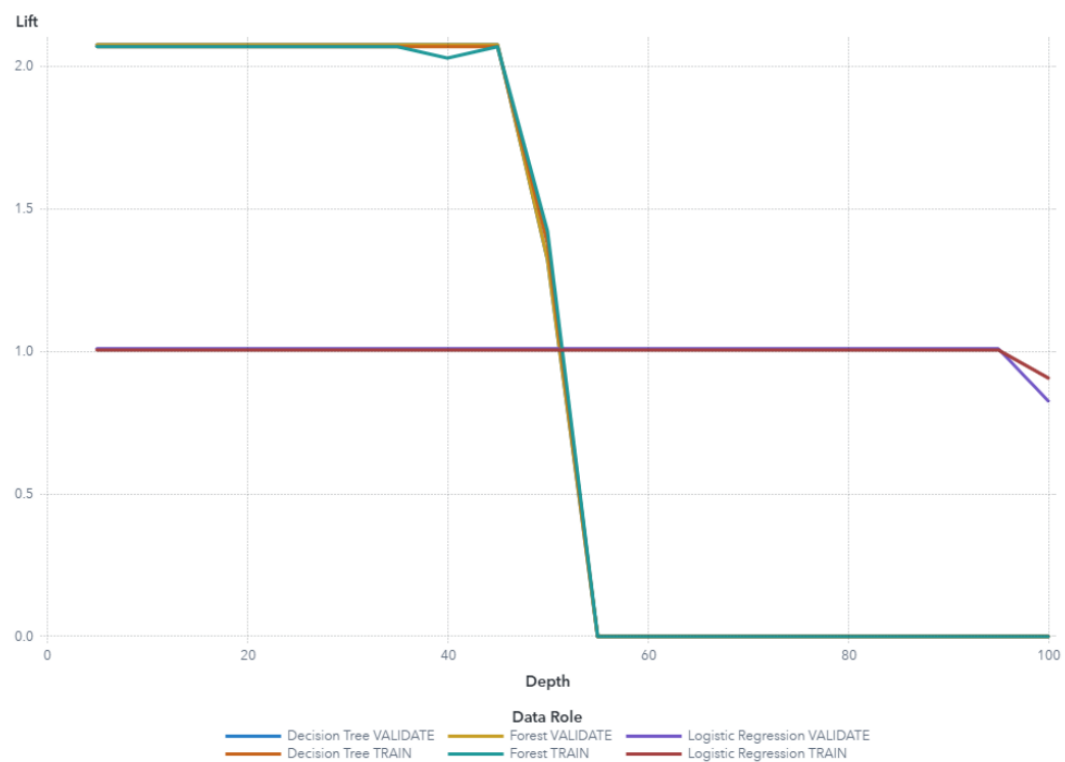
Multi-Class Log Loss	Misclassification at Cutoff	Misclassification Rate (Event)	Number of Observations	Root Average Squared Error
0.0000	0	0	436	0
0.8001	0.5138	0.4862	436	0.4189
0.0476	0	0	436	0.0637

Captured Response Percentage
10.3774
5.0459
10.3774

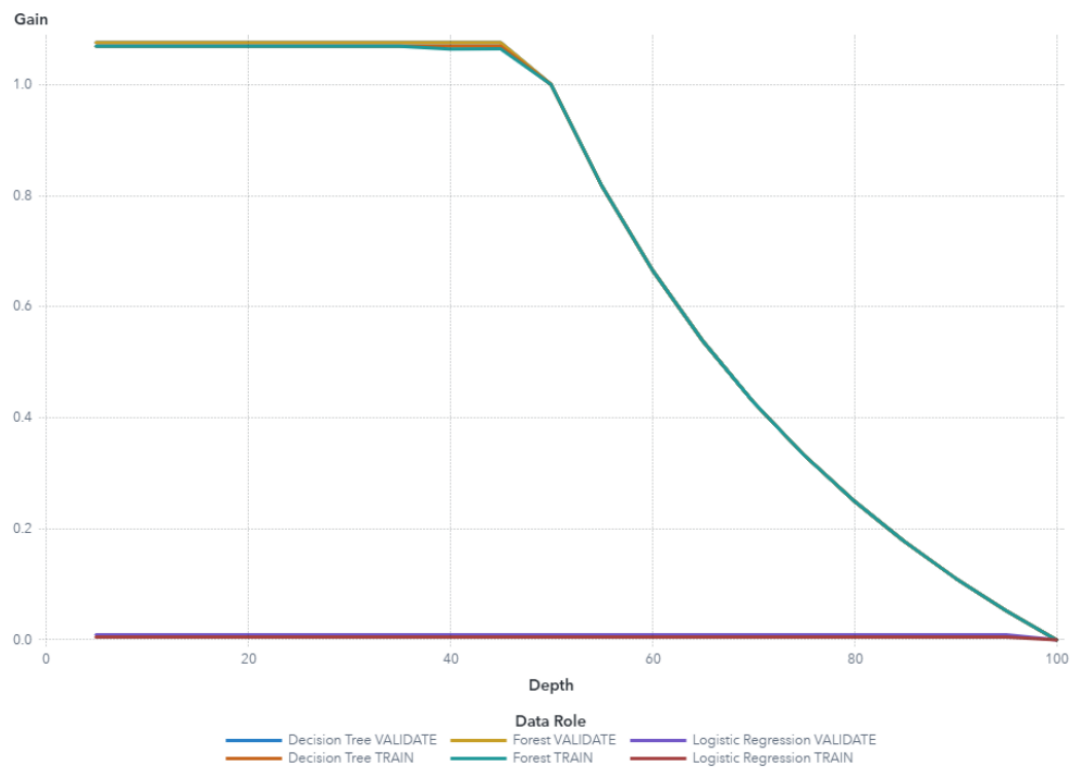
Cumulative Lift



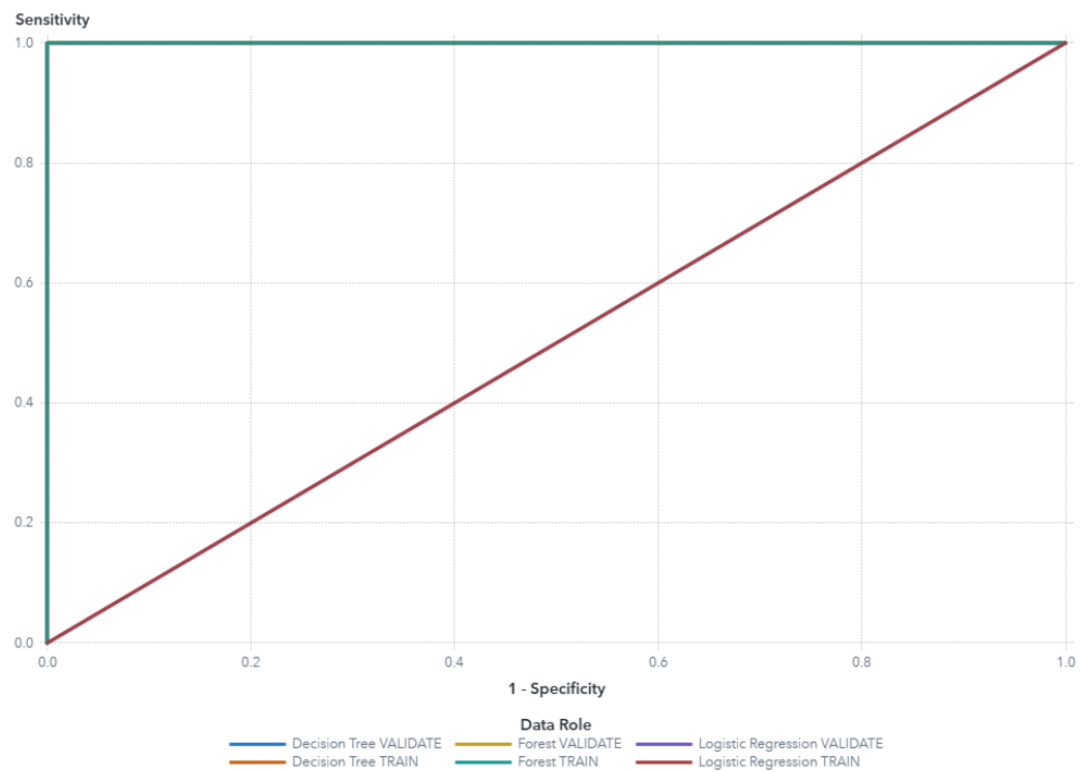
Lift



Gain



ROC



Accuracy

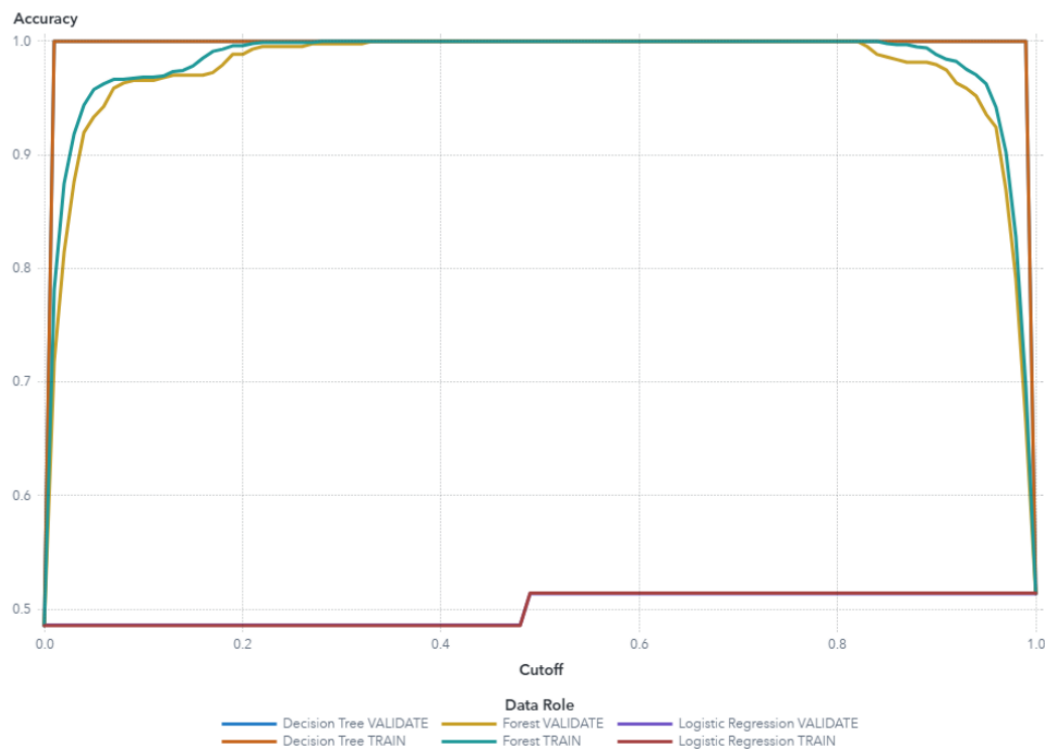


Figure 9: Comparison of predictive performance across classification models using KS statistic and accuracy on the validation dataset.

2.8 Interpretation and Road Safety Insights

The modelling results highlight that adjusted severity indicators and collision adjustment variables are the strongest predictors of collision severity. This suggests that post-collision assessments and situational adjustments capture critical information about accident seriousness.

The strong performance of tree-based models indicates complex, non-linear interactions between road conditions, environmental factors, and temporal features. From a road safety perspective, these findings support targeted interventions focusing on:

- High-risk road conditions
- Situations associated with severe outcomes
- Improved monitoring of severity indicators at accident scenes

2.9 Task 2 Summary

In Task 2, multiple predictive models were developed and evaluated to classify collision severity. The Decision Tree model was selected as the champion due to its superior performance and interpretability. The results provide robust predictive capability and valuable insights into the factors driving severe road traffic collisions, forming a strong foundation for further analysis and decision-making.

Task 3: Text Analytics of Road Traffic Related Tweets

3.1 Objective of Task 3

The objective of Task 3 is to apply text analytics techniques using SAS Visual Analytics and SAS Visual Text Analytics to explore unstructured Twitter data related to road traffic incidents. The analysis aims to identify dominant themes, key concepts, sentiment patterns, and topic structures within tweets, providing complementary qualitative insights to the structured accident data analysed in previous tasks.

3.2 Data Import and Preparation

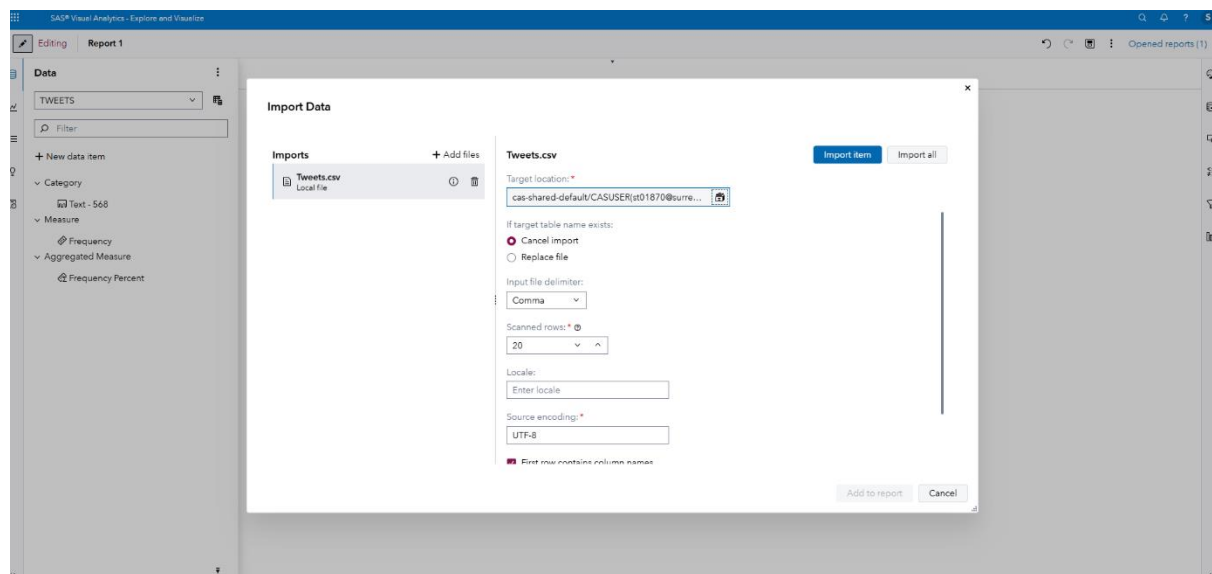


Figure 10: Import of Twitter dataset into SAS Visual Analytics and assignment of the Text variable for text analytics.

Text	Frequency	Frequency Percent
?? All 64s are delayed due to a collision at Farnham. The 1105 from Ipswich to Aldeburgh won't be able to run as a result ^RS	1	0.17%
?M25 is blocked both ways between J9 Leatherhead and J8 Reigate following a serious collision. Traffic is being held each way for LDNairamb to land. Expect long de...	1	0.17%
'You are just taking the mickey out of the British people by suggesting it was a work event'. [PM's head down] 'You know how silly that sounds don't you?' 'I repeat my...	1	0.17%
'#redhill #reigate @HarlequinTheat a lovely lady Liz helped me after my car accident last night but her number seems to have translated itself into someone else's ove...	1	0.17%
'Breaking News from@UKNIP247 The family of 19-year-old Christian Rowe from West End, Woking have today paid tribute to him following his death during a road tra...	1	0.17%
'Breaking News The family of 19-year-old Christian Rowe from West End, Woking have today paid tribute to him following his death during a road traffic collision on T...	1	0.17%
'Mount Lawley - collision in the left hand lane eastbound Guildford Rd at East Pde'	1	0.17%
'Road Traffic Collision: LEATHERHEAD BY-PASS jnc / HEADLEY ROAD Fri 11 Mar 1120 - 1135 involving a blue SUV and another vehicle. Witnesses or any dash cam fo...	1	0.17%
'Tarragindi - multi-vehicle collision southbound Pacific Motorway before Esher St'	1	0.17%
'There are long residual on the #M25 clockwise between J9 #Leatherhead and J10 #A3 #London following an earlier collision. All lanes are now open however journe...	1	0.17%
'UK driving licences of residents in Spain no longer valid from May 1st' Another #Brexit benefit! This just reminds me of a driving licence story of mine. Years ago I ha...	1	0.17%
'Wednesday 9th March 2022 @ 17:10 M25 Accident Report M3 eastbound between junctions J3 (Woking) & J2 (Gatwick M23) Lane one is closed Road traffic collision...	1	0.17%
(SurreyLive):#M25 collision closes lanes near #Reigate and Leatherhead: Two lanes were closed on the M25 clockwise near Reigate and Leatherhead following a road ...	1	0.17%
(SurreyLive):#Man arrested at #Guildford railway station after crash in Normandy required 'vital first aid': Officers say arrest followed previous road traffic collision wh...	1	0.17%
@AM730Traffic accident in Surrey/delta by serpentine river 91/99 connector is completely stopped. 4/5 car accident one car completely flipped over. Hopefully every...	1	0.17%
@CrispinBlunt you are er, umm, ahh trending and it ain't for a good reason. Utter car crash performance tonight, blindly supporting the PM. But an 18,000 majority ap...	1	0.17%
@dt808traffic multiple car accident Redhill eastbound (Twitter link not provided)	1	0.17%
@SurreyPolice are appealing for witnesses following a devastating fatal collision on the #Farnham bypass in the early hours of this morning if you able to assist please...	1	0.17%
#A3 northbound between #A283 near #Milford (north) and #B3000 near #Compton - approaching #Guildford. 1 lane (of 2) is closed due to a collision. Injuries and de...	1	0.17%
#BF_BUS_405 London Road North / Merstham: Route 405 is operating between Croydon and Coulsdon Town only due to a collision. Stops from Coulsdon Town to Re...	1	0.17%
#BF_BUS_405 London Road North / Merstham: Route 405 is operating between Croydon and Hooley only due to a collision. Stops from Church Lane to Redhill Bus St...	1	0.17%
#BirdyBike at last, replaced the bike written off in the 2018 collision. Very comfy ride reacquainting myself with this kind of bike, around Woking (bike shop in Chobha...	1	0.17%
#BROADBRIDGEHEATH A281 GUILDFORD ROAD BOTH DIRECTIONS BLOCKED between Farlington School and A29 Bognor Road, due to earlier collision.	1	0.17%
#Brockham: #A25 Reigate Rd is closed btwn Brockham Lane and B2032 Station Rd for emergency repairs after a serious collision. Traffic btwn Reigate and Dorking is a...	1	0.17%
#Cobham: #A245 Byfleet Rd is partly blocked between B365 Seven Hills Rd and Redhill Rd after a collision.	1	0.17%
#Exercise Heron @Hampton court. Organised by Surrey FRS the exercise scenario was the simulation of a collision between two vessel mid channel on the river Tham...	1	0.17%
#FARNHAM A325 WEST STREET PARTIALLY BLOCKED BOTH DIRECTIONS queueing traffic due to collision between A287 Downing Street and Crondall Lane.	1	0.17%
#GODALMING B2130 THE STREET BOTH DIRECTIONS Reports of slow traffic due to collision between Munstead Heath Road (Godalming) and Thorncombe Street (H...	1	0.17%
#GUILDFORD A3 SOUTHBOUND Reports of queueing traffic due to collision on A3 Southbound at Egerton Road (Cathedral Turn).	1	0.17%

Figure 11: A dataset containing 598 tweets was imported into SAS Visual Analytics from a CSV file (Tweets.csv).

The dataset consists of two variables:

- `_uniqueid_`: numeric identifier (Key)
- Text: tweet content (unstructured text)

The text variable was explicitly assigned the Text role, which is required for text parsing and downstream text analytics pipelines.

3.3 Exploratory Text Visualisation (Word Cloud)

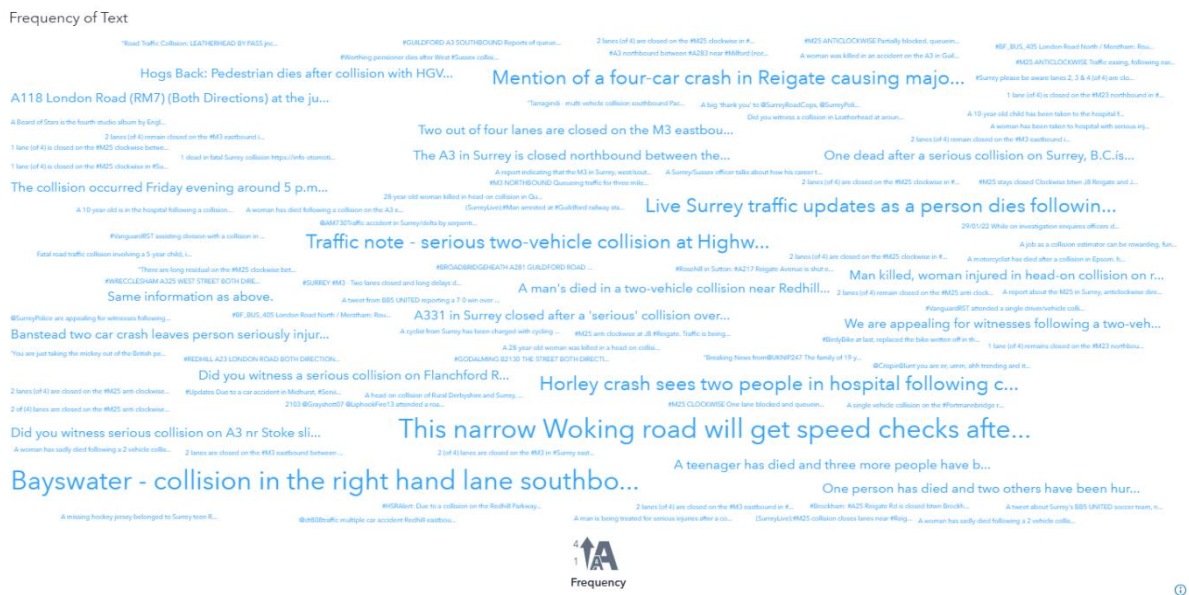


Figure 12: Word cloud showing the most frequently occurring terms in road traffic-related tweets.

An initial word cloud was created to visually explore the most frequent terms appearing across the tweets.

The size of each word reflects its relative frequency, allowing rapid identification of commonly discussed events.

Key recurring terms include:

- collision
- road
- traffic
- closed
- serious
- accident

These terms confirm that the dataset is strongly focused on road traffic incidents, validating its relevance to the study

3.4 Text Analytics Pipeline Design

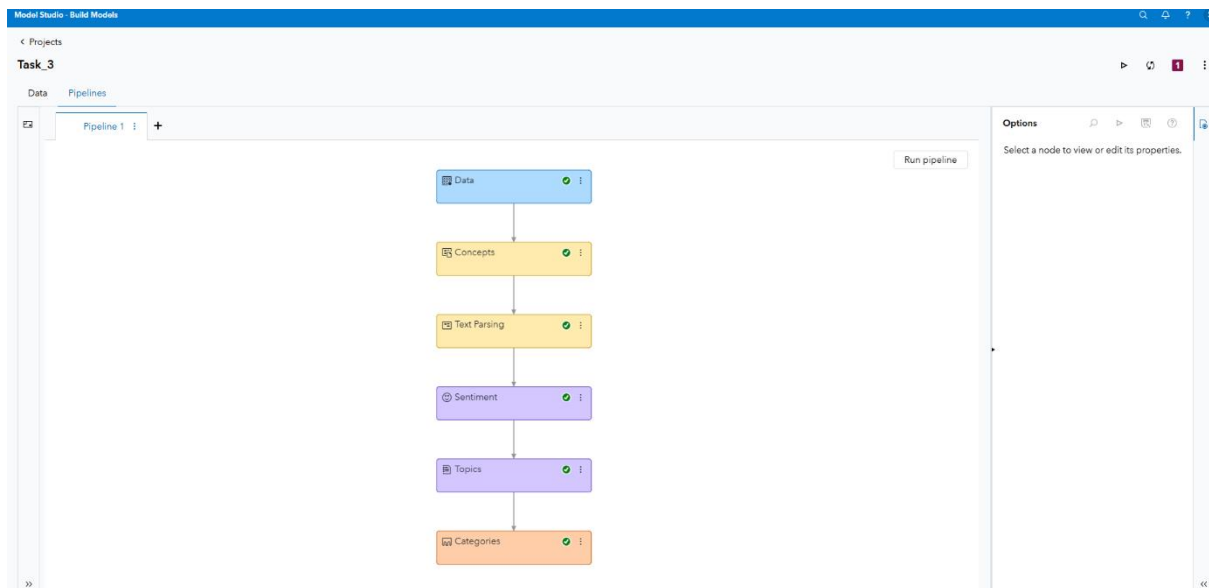


Figure 13: SAS Visual Text Analytics pipeline illustrating the sequential application of Concepts, Text Parsing, Sentiment, Topics, and Categories nodes.

A structured text analytics pipeline was built in SAS Model Studio, consisting of the following sequential nodes:

1. Concepts – to detect predefined and custom concepts
2. Text Parsing – to tokenize, stem, and filter text
3. Sentiment – to classify tweets as positive, negative, mixed, or neutral
4. Topics – to identify latent themes across tweets
5. Categories – to group documents into interpretable traffic-related categories

This pipeline ensures that both linguistic structure and semantic meaning are captured.

3.5 Concept Extraction Results

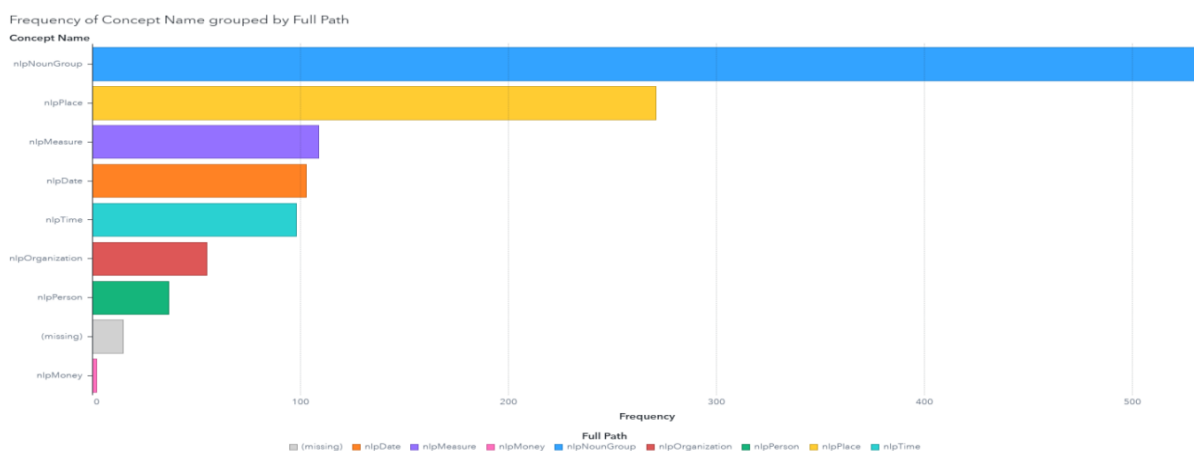


Figure 14: Extracted traffic-related concepts identified from tweet text using the Concepts node.

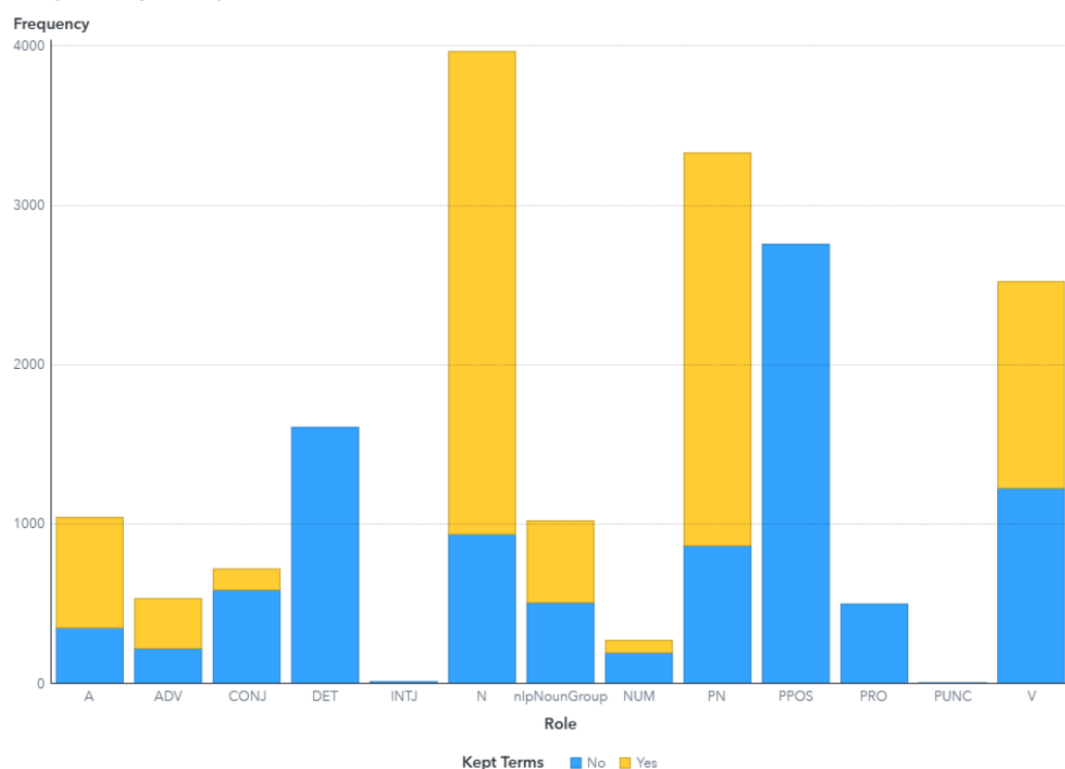
The Concepts node identified key traffic-related entities and phrases, including:

- Road names (e.g., A3, M25)
- Incident descriptors (e.g., serious collision, lane closed)
- Emergency and disruption indicators

These concepts provide structured signals that help interpret tweet content beyond raw word frequency. (Appendix D4)

3.6 Text Parsing and Linguistic Processing

Role by Frequency



Descriptive Statistics

Measure	Terms in a Sentence	Terms in a Document
Minimum	1	1
Maximum	49	65
Mean	14.0932	30.5903

Figure 15: Text parsing output showing tokenisation and filtering of tweet content.

The Text Parsing node performed tokenisation, stemming, and stop-word removal to standardise the text data.

This step reduced noise (e.g., URLs, punctuation, common stop words) and improved the quality of downstream sentiment and topic modelling. (Appendix D5)

3.7 Sentiment Analysis

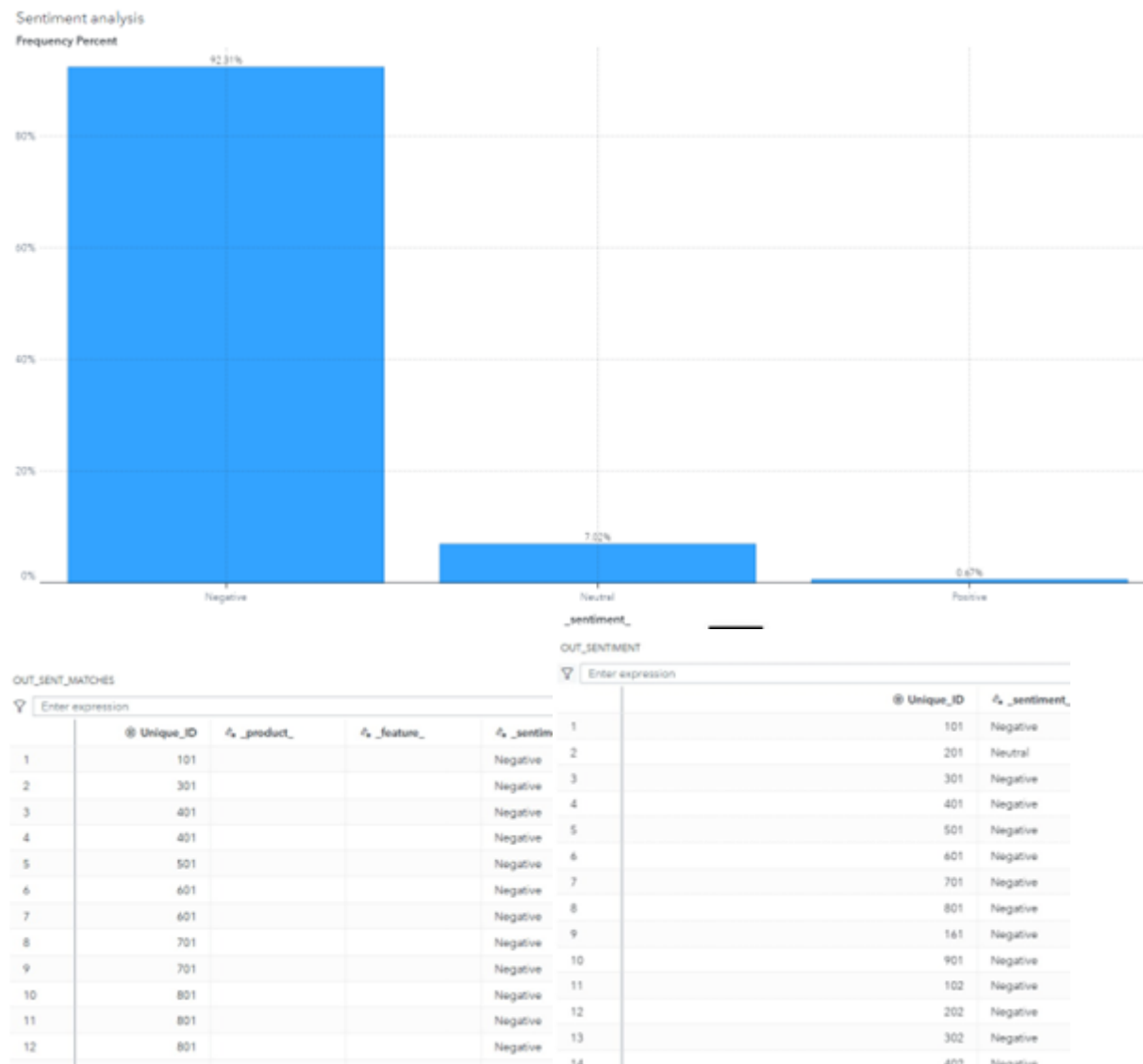


Figure 16: Sentiment distribution of traffic-related tweets, showing dominance of negative sentiment.

The Sentiment node classified tweets into four categories:

- Negative
- Positive
- Mixed
- No sentiment

The results indicate a strong dominance of negative sentiment, which is expected given the focus on collisions, road closures, and fatalities. 92% of the tweets are composed of negative sentiments. The reason for this can be that the tweets are related to accidents, collision, block etc. 7% tweets have neutral sentiment while only 0.67% has positive sentiments. Positive sentiment appears rarely and is usually associated with recovery updates or reopening of roads. (Appendix D6)

3.8 Topic Modelling Results

Number of Documents Per Topic

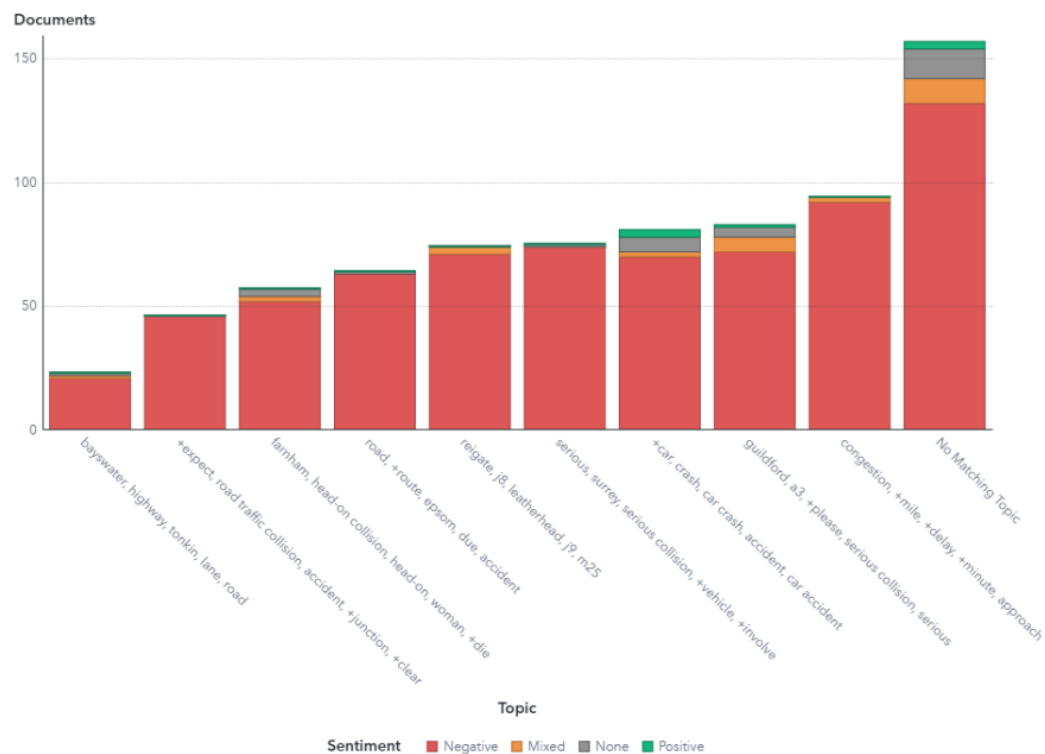


Figure 17: Number of documents per topic with sentiment overlay, highlighting dominant traffic incident themes.

The Topics node identified multiple recurring themes within the tweets, including:

- Serious multi-vehicle collisions
- Road closures and congestion
- Emergency response and delays
- Unassigned or miscellaneous traffic updates

Approximately 26% of documents were not assigned to a specific topic, indicating either short or ambiguous tweets. (Appendix D7)

3.9 Category Classification



Figure 18: Tweet categorisation into major traffic-related themes using the Categories node.

The Categories node grouped tweets into higher-level, interpretable categories such as:

- Serious collisions
- Guildford Roads

This categorisation enhances interpretability and allows policymakers or traffic authorities to quickly understand operational impacts discussed on social media. (Appendix D8)

3.10 Summary

Task 3 demonstrates that text analytics can effectively extract actionable insights from unstructured social media data. The analysis reveals that tweets predominantly focus on serious collisions, congestion, and road closures, with sentiment strongly skewed toward negative outcomes. These findings complement the structured accident modelling results from Task 2 and provide additional contextual and real-time perspectives on road traffic incidents.

Task 4 – Decision-Maker’s Summary

Purpose

This study analysed UK road traffic accident data alongside social media content to identify key factors influencing accident severity and to understand public perception of road traffic incidents. The analysis combines structured accident records with unstructured Twitter data to support evidence-based road safety decision-making.

Key Findings

1. Factors Driving Collision Severity

Predictive modelling results show that adjusted collision severity indicators, road characteristics, and situational conditions are the strongest predictors of accident severity. Tree-based models consistently outperformed traditional statistical models, indicating that non-linear interactions between road, environmental, and temporal factors play a critical role in determining accident outcomes.

Implication:

Road safety interventions should prioritise locations and situations where combinations of risk factors are present, rather than focusing on single variables in isolation.

2. Model Performance and Reliability

Among the models tested, the Decision Tree model delivered the highest predictive performance while remaining interpretable. Its transparent structure makes it suitable for operational use, enabling stakeholders to understand why certain collisions are classified as high risk.

Implication:

Decision Trees can be confidently used to support risk assessment tools for transport authorities and emergency planning teams.

3. Temporal and Spatial Risk Patterns

Feature engineering revealed that time of day, calendar month, and geographic location significantly influence accident patterns. Higher-risk periods are associated with peak travel hours and specific seasonal conditions.

Implication:

Targeted enforcement, public awareness campaigns, and resource allocation should be aligned with high-risk times and locations, rather than applied uniformly.

4. Public Sentiment and Perception

Text analytics of traffic-related tweets shows a strong dominance of negative sentiment, reflecting public concern around collisions, injuries, road closures, and travel disruption. Topic and category analysis confirms that serious accidents and congestion are the most discussed themes.

Implication:

Public communication strategies should focus on timely updates, transparency, and reassurance, particularly during major incidents, to manage public anxiety and improve trust.

Strategic Recommendations

1. Adopt predictive risk screening - Use interpretable machine learning models (e.g. Decision Trees) to identify high-risk collision scenarios proactively.
 2. Target interventions by time and location - Focus enforcement, infrastructure improvements, and emergency readiness on periods and areas with elevated risk.
 3. Integrate social media monitoring - Combine accident data with real-time sentiment analysis to detect emerging issues and public concerns faster.
 4. Improve public communication during incidents - Address the strong negative sentiment by providing clear, frequent updates during road traffic disruptions.
-

Overall Conclusion

By integrating structured accident data with unstructured social media insights, this study provides a comprehensive view of road traffic risk and public response. The findings support more targeted, data-driven, and transparent road safety strategies, enabling policymakers and transport authorities to reduce accident severity and improve public confidence.

References

- Aggarwal, C.C. and Zhai, C. (2012) Mining text data. Boston, MA: Springer.
- Blei, D.M., Ng, A.Y. and Jordan, M.I. (2003) 'Latent Dirichlet Allocation', Journal of Machine Learning Research, 3, pp. 993–1022.
- Pang, B. and Lee, L. (2008) 'Opinion mining and sentiment analysis', Foundations and Trends® in Information Retrieval, 2(1–2), pp. 1–135.
- SAS Institute Inc. (2023) Exploring SAS® Viya®: data mining and machine learning. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) Exploring SAS® Viya®: visual analytics and statistics. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) Text analytics with SAS®. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) SAS® Visual Analytics: user's guide. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) SAS® Visual Data Mining and Machine Learning: user's guide. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) SAS® Visual Data Mining and Machine Learning: reference. Cary, NC: SAS Institute Inc.
- SAS Institute Inc. (2023) SAS® Visual Text Analytics: user's guide. Cary, NC: SAS Institute Inc.
- Witten, I.H., Frank, E., Hall, M.A. and Pal, C.J. (2016) Data mining: practical machine learning tools and techniques. 4th edn. Cambridge, MA: Morgan Kaufmann.

Appendix – A: Task 0 – Random Sample Selection

A1: Task 0 Sampling Workflow (SAS Viya Flow)

The screenshot displays the SAS Viya Flow interface. At the top, there's a navigation bar with tabs for 'Start Page', 'Task_0.fiw', 'Task_1.fiw', 'Task_1_Cleaning.fiw', 'Task_1_Balancing.fiw', and 'SAS Program.sas'. Below this is a toolbar with various icons for running, canceling, and viewing the flow. The main area shows a flow diagram with nodes: 'RoadData_2' (input), 'Import File' (process), 'Road' (dataset), 'Select Random...' (process), and 'Road_3500' (output). Below the flow diagram, the 'Road_3500' dataset is selected, and its properties are shown. The 'Table Properties' tab is active, displaying a preview of the data. The table has 3500 rows and 44 columns. The first five rows are shown, with columns: 'collision_index', 'collision_year', 'collision_ref_no', 'location_easting_osgr', and 'location_northing_osgr'.

	@ collision_index	@ collision_year	@ collision_ref_no	@ location_easting_osgr	@ location_northing_osgr
1	2.0234613E12	2023	461346475	580398	166308
2	2.0234613E12	2023	461274241	623262	136637
3	2.0234613E12	2023	461295594	568793	173097
4	2.0234613E12	2023	461289687	599534	142971
5	2.0234613E12	2023	461300536	565084	173680

Caption: SAS Viya flow used to import the road traffic accident dataset and generate a random sample of 3,500 observations.

A2: Output Dataset Confirmation

The screenshot shows the 'Results:Task_0.fiw' window. It displays the 'Selection Method' as 'Simple Random Sampling'. Below this, a table shows the 'Input Data Set' as 'ROAD', 'Random Number Seed' as '6859062', 'Sample Size' as '3500', 'Selection Probability' as '0.898357', 'Sampling Weight' as '1.113143', and 'Output Data Set' as 'ROAD_3500'. Below the table, a preview of the 'WORK.Road_3500' dataset is shown. The table has 3500 rows and 14 columns: 'Obs', 'collision_index', 'collision_year', 'collision_ref_no', 'location_easting_osgr', 'location_northing_osgr', 'longitude', 'latitude', 'police_force', 'collision_severity', 'number_of_vehicles', 'number_of_casual', 'date', and 'day_of_week'.

Obs	collision_index	collision_year	collision_ref_no	location_easting_osgr	location_northing_osgr	longitude	latitude	police_force	collision_severity	number_of_vehicles	number_of_casual	date	day_of_week
1	2.0234613E12	2023	461346475	580398	166308	0.990123	51.367182	46	3	4	1	07/14/2023	1
2	2.0234613E12	2023	461274241	623262	136637	1.186491	51.085582	46	3	2	1	02/05/2023	1
3	2.0234613E12	2023	461295594	568793	173097	0.426804	51.431736	46	2	2	1	03/17/2023	1
4	2.0234613E12	2023	461289687	599534	142971	0.851666	51.151206	46	3	2	2	03/19/2023	1
5	2.0234613E12	2023	461300536	565084	173680	0.373769	51.438066	46	3	1	1	04/17/2023	1
6	2.0234613E12	2023	461303292	56432	139363	0.234375	51.132184	46	3	2	1	04/22/2023	1
7	2.0234613E12	2023	461322647	561721	168549	0.323088	51.392937	46	3	2	2	05/22/2023	1
8	2.0234613E12	2023	461330018	630197	144654	1.280454	51.154814	46	3	1	1	07/03/2023	1
9	2.0234613E12	2023	461339482	578359	152610	0.55402	51.244779	46	3	2	5	07/30/2023	1
10	2.0234613E12	2023	461347463	559406	148662	0.280965	51.214915	46	2	2	1	08/29/2023	1
11	2.0234614E12	2023	461356966	627499	143308	1.25109	51.143808	46	3	1	1	09/17/2023	1
12	2.0234614E12	2023	461367266	591845	164215	0.753239	51.344636	46	3	1	1	10/20/2023	1
13	2.0234614E12	2023	461391613	622323	149209	1.18086	51.19882	46	3	1	2	12/23/2023	1
14	2.0234613E12	2023	461330025	575659	163901	0.520929	51.347044	46	3	1	1	07/10/2023	1
15	2.0234613E12	2023	461332616	580442	167142	0.591177	51.374659	46	2	2	1	07/12/2023	1
16	2.0234614E12	2023	461359788	577728	168847	0.563077	51.39083	46	3	2	1	09/25/2023	1
17	2.0234613E12	2023	461283388	599546	141178	0.850844	51.135099	46	3	2	1	03/01/2023	1
18	2.0234613E12	2023	461327699	565442	172380	0.378308	51.426282	46	3	2	1	07/04/2023	1
19	2.0234613E12	2023	461346710	563300	173302	0.34795	51.435186	46	3	2	1	07/31/2023	1
20	2.0234614E12	2023	461379536	565374	171685	0.377007	51.420058	46	2	1	1	10/24/2023	1
21	2.0234613E12	2023	461396950	555305	167955	0.230685	51.389391	46	3	3	4	03/04/2023	1
22	2.0234613E12	2023	461327856	550555	158851	0.158622	51.308868	46	2	2	1	06/20/2023	1
23	2.0234613E12	2023	461316166	613750	133404	1.048973	51.060172	46	3	1	1	05/20/2023	1
24	2.0234613E12	2023	461263867	594384	172632	0.794252	51.419367	46	3	2	1	01/08/2023	1
25	2.0234613E12	2023	461347644	594321	172364	0.7932	51.416981	46	3	1	1	08/19/2023	1
26	2.0234614E12	2023	461368650	603061	159879	0.924493	51.301484	46	3	2	1	10/24/2023	1
27	2.0234614E12	2023	461373580	575554	161943	0.518458	51.329488	46	3	2	1	10/31/2023	1
28	2.0234614E12	2023	461394176	575446	170461	0.52111	51.406036	46	3	2	1	12/06/2023	1
29	2.0234613E12	2023	461261788	601829	161745	0.895006	51.319	46	2	2	1	01/05/2023	1
30	2.0234613E12	2023	461269516	565634	144701	0.36823	51.177557	46	3	1	1	01/15/2023	1
31	2.0234613E12	2023	461268102	593920	150238	0.775415	51.218402	46	3	2	2	01/19/2023	1
32	2.0234613E12	2023	461310830	630503	164766	1.307761	51.335235	46	3	2	1	05/11/2023	1

Output dataset (WORK.Road_3500) confirming 3,500 observations and 44 variables after random sampling.

A3: SAS Log Evidence

```
02/01/2026, 17:57 SAS Log

1  /*=====*/
2  * Flow:      Task_0.flw
3  * ID:
4  * Created:    2026-01-02T17:56:00.446323575Z
5  * Created by: st01870@surrey.ac.uk
6  * Modified:   2026-01-02T17:56:00.446323575Z
7  * Modified by: st01870@surrey.ac.uk
8  *
9  * Output Tables:
10 * WORK.Road
11 * WORK.Road_3500
12 *
13 * Version:     DataFlows lts 2025.03 (20251010.1760100479962)
14 *
15 * Generated On: 2026-01-02T17:56:00.446323575Z
16 * Generated by: st01870@surrey.ac.uk
17 *=====*/
18 /* region: Generated preamble */
19
20 /* Make sure the current directory is writable */
21 data _null_;
22   length rc 4;
23   %let tworkloc=%sysfunc(getoption(work));
24   rc=dlgcdir(&tworkloc);
25 run;
NOTE: The current working directory is now
"/opt/sas/viya/config/var/tmp/compsrv/default/78023247-1c0f-4e5b-8278-796d3e4ddde8/SAS_work83240000211_sas-compute-server-726
b3cc4-8c35-4ae3-95b7-e1782431d04f-42492".
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds

26
27 /* Setup options */
28 title;
29 footnote;
30 options validvarname=any;
31 options validmemname=extend;
32 options dtrset date number;
33 options device=png;
34
35 /* Setup macro variables */
36 %let syscc=0;
37 %let _clientapp = %nrquote(%nrstr(SAS Studio));
38 %let _clientappabbrev = %nrquote(%nrstr(Studio));
39 %let _clientappversion=2025.03;
40 %let _clientversion=6.1072.10;

https://vfl-040.engage.sas.com/SASStudio/ 1/20
```

SAS log confirming successful execution of the random sampling procedure using a fixed seed and sample size.

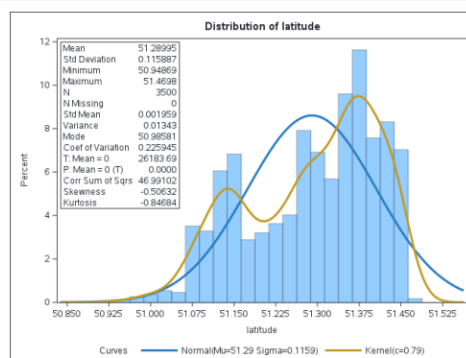
Appendix - B: Task 1- Data Exploration and Analysis

B1: Raw Dataset and Structure

02/01/2026, 16:59

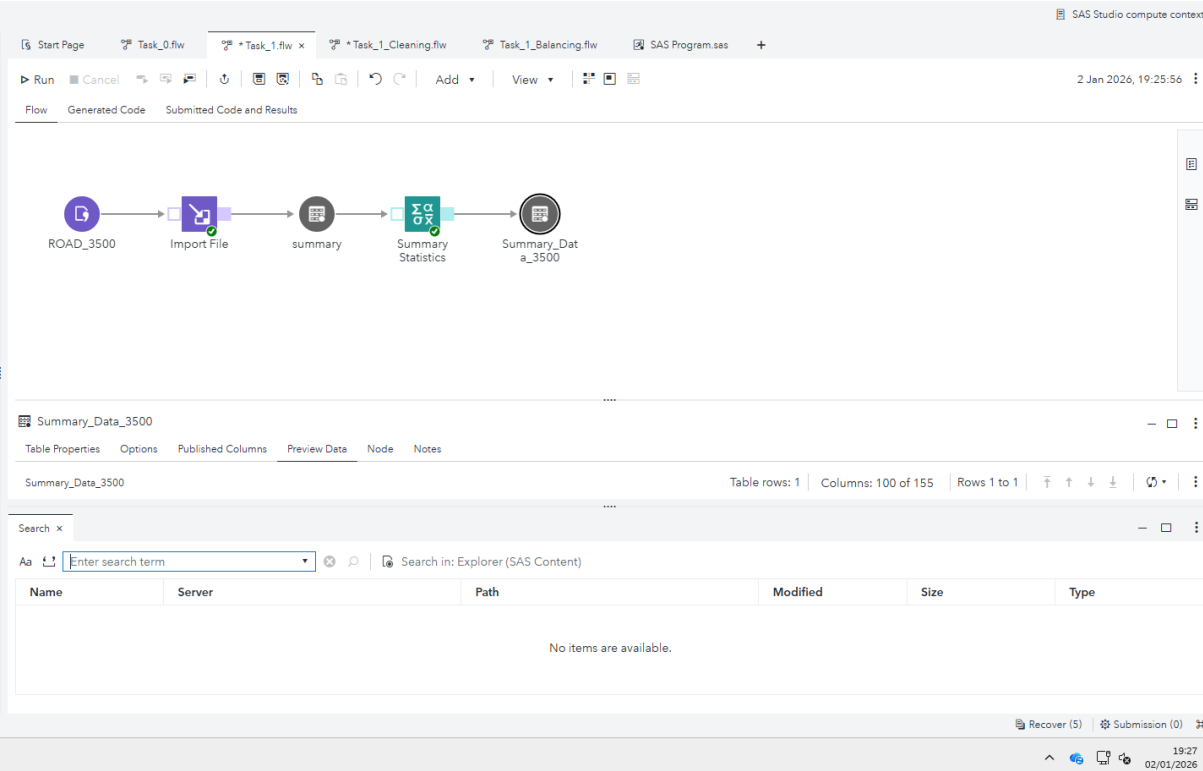
Results:Task_1.flw

Variable	Label	Mean	Std Dev	Minimum	Maximum	N	N Miss	Std Error	Variance	Mode	Lower 95% CL for Mean	Upper 95% CL for Mean	Coeff of Variation	t Value	Pr > t	Corrected SS	Skewness
latitude	latitude	51.2899481	0.1158872	50.9486900	51.4699040	3500	0	0.0019589	0.0134298	50.9858060	51.2861075	51.2937887	0.2259452	26183.7	<.0001	46.9910202	-0.5063216
longitude	longitude	0.6798734	0.3825770	0.0407980	1.4451940	3500	0	0.0064687	0.1463851	0.1784710	0.6671944	0.6925523	58.2717979	105.13	<.0001	512.1316227	0.4941869
number_of_vehicles	number_of_vehicles	1.8482857	0.7563845	1.0000000	9.0000000	3500	0	0.0128190	0.5751471	2.0000000	1.8734193	41.0317804	144.18	<.0001	2012.44	1.7669408	
number_of_casual	number_of_casual	1.3543217	0.7759906	1.0000000	9.0000000	3494	6	0.0131279	0.6021614	1.0000000	1.3285826	1.3800608	57.2973608	103.16	<.0001	2103.35	3.1330599
speed_limit	speed_limit	40.7257143	15.0843744	20.0000000	70.0000000	3500	0	0.2549725	227.5383497	30.0000000	40.2258045	41.2256241	37.0389436	159.73	<.0001	796156.69	0.8279259
weather_cond	weather_cond	1.4034286	1.3471848	1.0000000	9.0000000	3500	0	0.0227716	1.8149068	1.0000000	1.3587817	1.4480755	95.9923998	61.63	<.0001	8350.36	4.4069568
collision_severity	collision_severity	2.7750143	0.4448209	1.0000000	3.0000000	3498	2	0.0075210	0.1978657	3.0000000	2.7602683	2.7897603	16.0295009	368.97	<.0001	691.9362493	-1.7103894
date	date	23195.06	104.5706228	23011.00	23375.00	3500	0	1.7675661	10935.02	23058.00	23191.60	23198.53	0.4508314	13122.6	<.0001	38281618.04	-0.0723466
light_cond	light_cond	1.9571429	1.7432818	1.0000000	7.0000000	3500	0	0.0294668	3.0390316	1.0000000	1.8993689	2.0149168	89.0727945	66.42	<.0001	10633.57	1.5217724



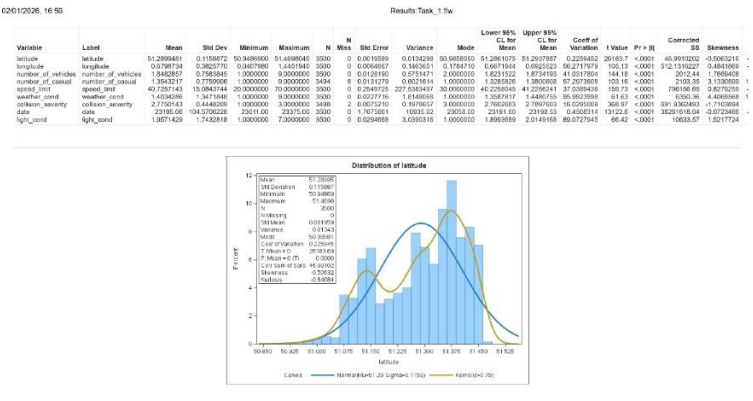
Preview and structural overview of the sampled road traffic accident dataset, showing variable names, data types, and initial dataset dimensions prior to cleaning and feature engineering.

B2: Summary Statistics

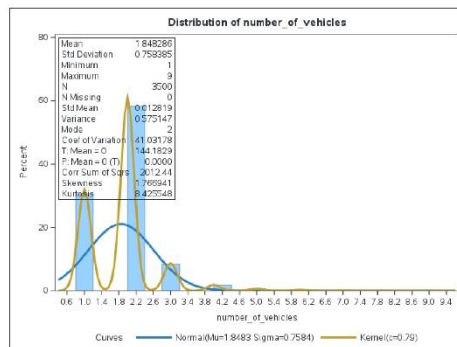
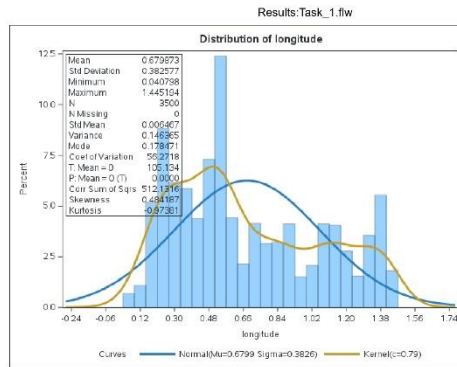


Summary statistics of key numerical and categorical variables used to support exploratory data analysis and identify central tendencies and variability within the dataset.

B3: Variable Distributions

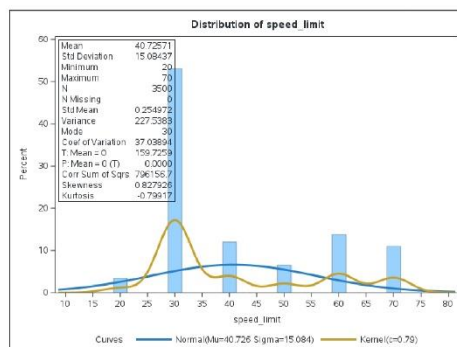
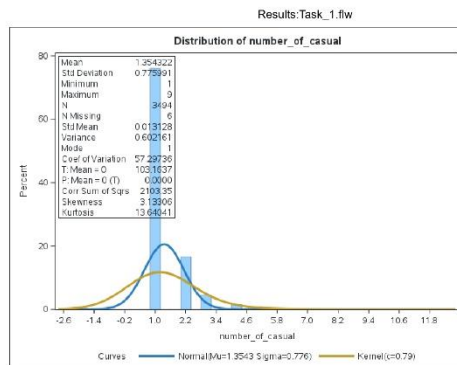


02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

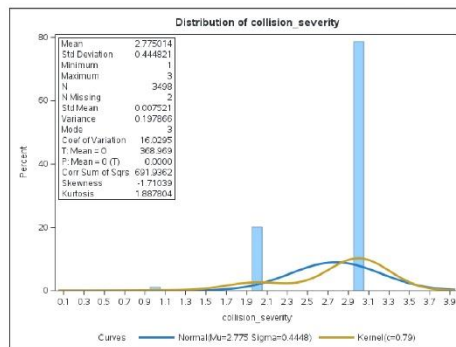
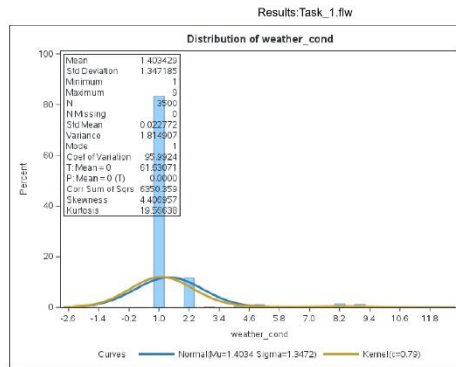
2/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

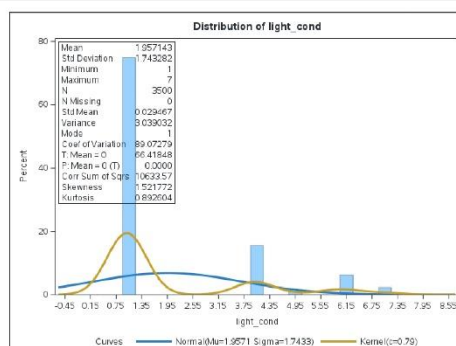
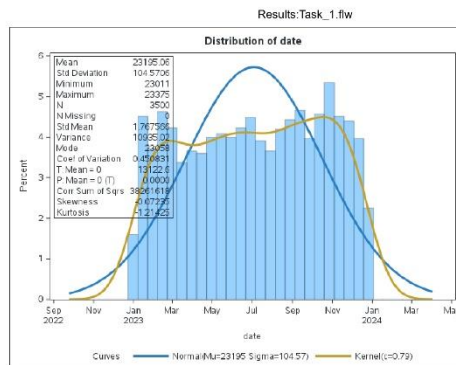
3/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

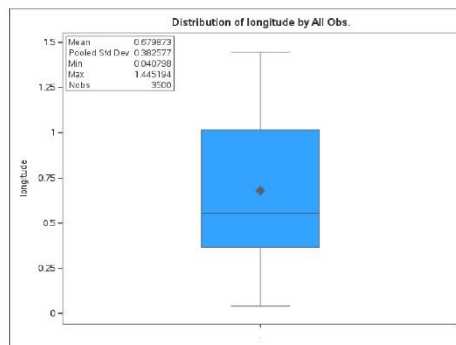
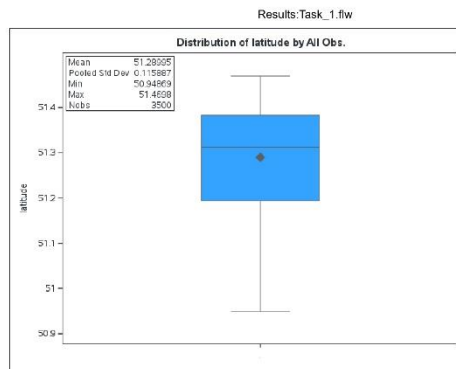
4/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

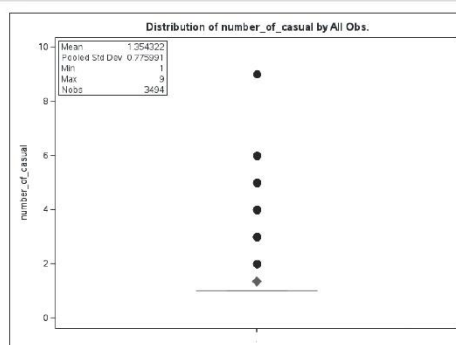
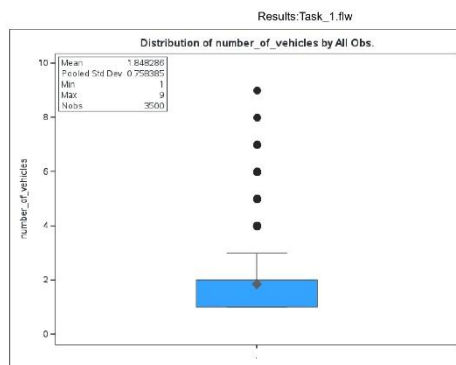
5/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

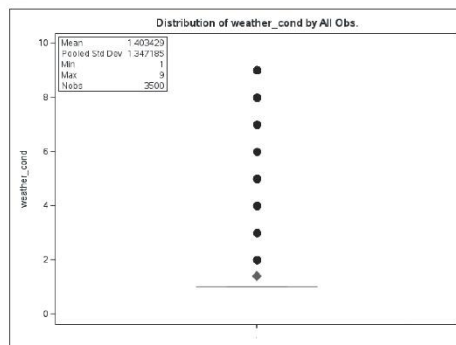
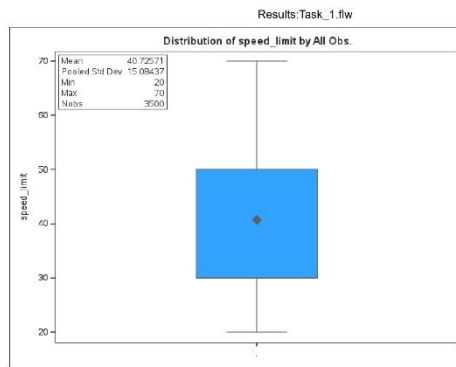
6/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

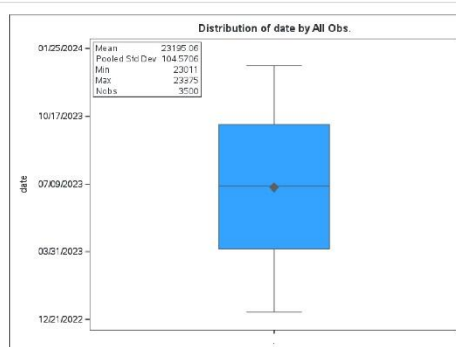
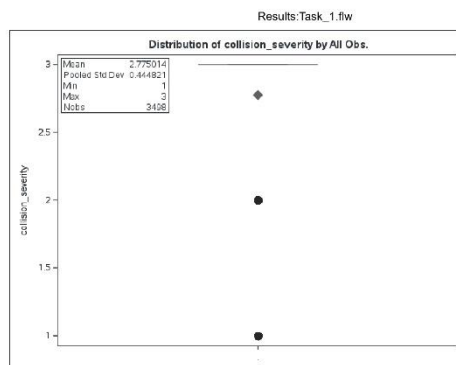
7/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

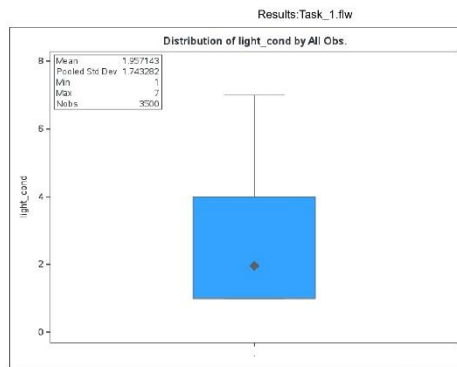
8/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

9/10

02/01/2026, 16:59

<https://vfl-040.engage.sas.com/SASStudio/>

10/10

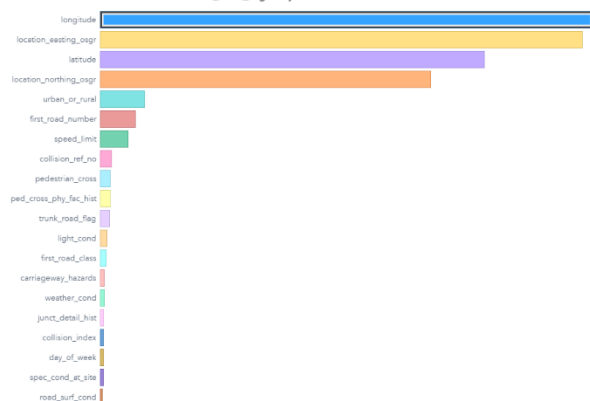
Distribution plots of key variables, illustrating data spread, skewness, and potential outliers identified during exploratory data analysis.

B4: Local Authority Highway Analysis

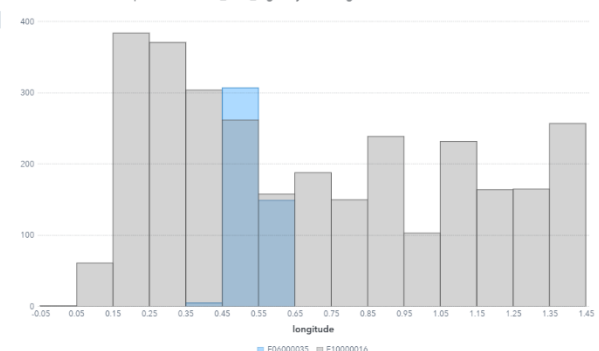
What are the characteristics of local_auth_highway?

E06000035 is less common at 13.17% (461 of 3.5K). E10000016 is more common at 86.83%. The three most related factors are longitude, location_easting_osgr, and latitude.

What factors are most related to local_auth_highway?



What is the relationship between local_auth_highway and longitude?

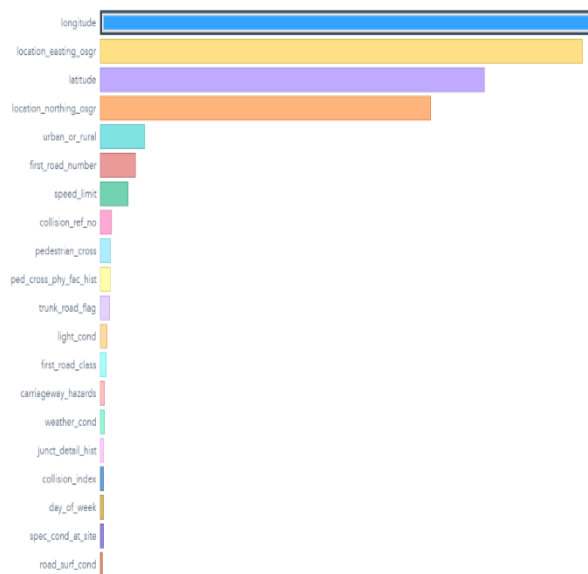


The average longitude is 0.53 when local_auth_highway is E06000035, with a minimum of 0.44 and a maximum of 0.65. The average longitude is 0.7 when local_auth_highway is E10000016, with a minimum of 0.04 and a maximum of 1.4. Average longitude is 0.68, and it ranges from 0.04 to 1.4.

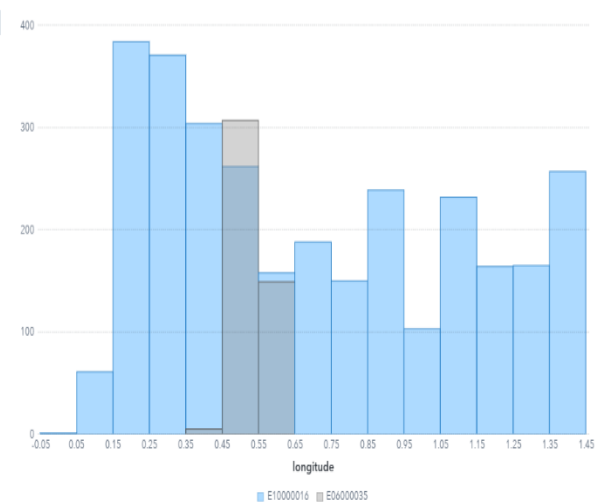
What are the characteristics of local_auth_highway?

E10000016 is more common at 86.83% (3K of 3.5K). E06000035 is less common at 13.17%. The three most related factors are longitude, location_easting_osgr, and latitude.

What factors are most related to local_auth_highway?



What is the relationship between local_auth_highway and longitude?



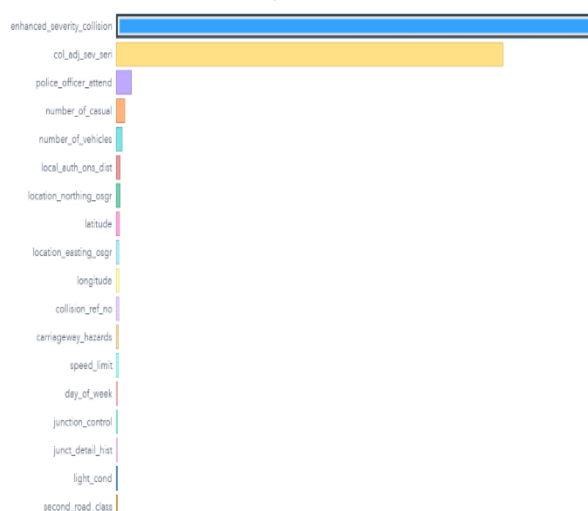
The average longitude is 0.7 when local_auth_highway is E10000016, with a minimum of 0.04 and a maximum of 1.4. The average longitude is 0.53 when local_auth_highway is E06000035, with a minimum of 0.44 and a maximum of 0.65. Average longitude is 0.68, and it ranges from 0.04 to 1.4.

What are the characteristics of collision_severity?

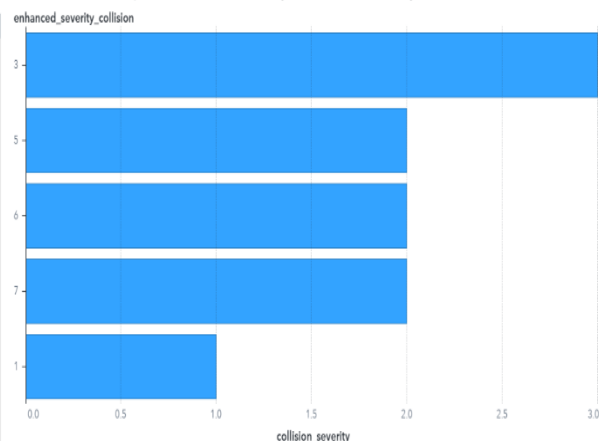
collision_severity ranges from 1 to 3. Average collision_severity is 2.8. Most cases (the middle 80%) have collision_severity between 2 and 3. col_adj_sev_seri best differentiates the highest (top 10%) and the lowest (bottom 10%) collision_severity cases. The three most related factors are enhanced_severity_collision, col_adj_sev_seri, and police_officer_atten.

d. There are 746 cases that might be outliers, with collision_severity less than or equal to 2.

What factors are most related to collision_severity?



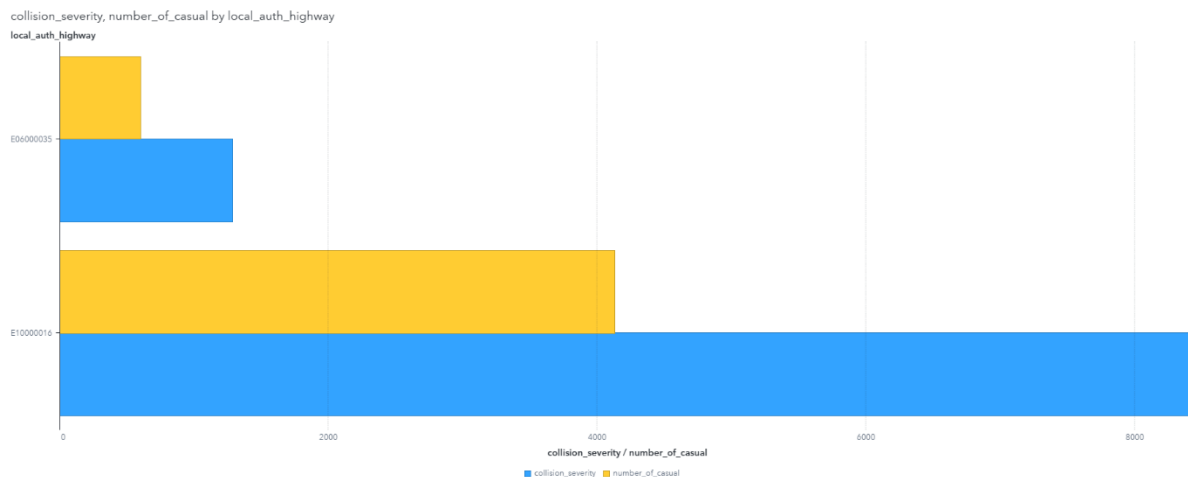
What is the relationship between collision_severity and enhanced_severity_collision?



When enhanced_severity_collision is 3, the average of collision_severity is a high value. When enhanced_severity_collision is 1, the average of collision_severity is a low value. The most common enhanced_severity_collision value is 3.

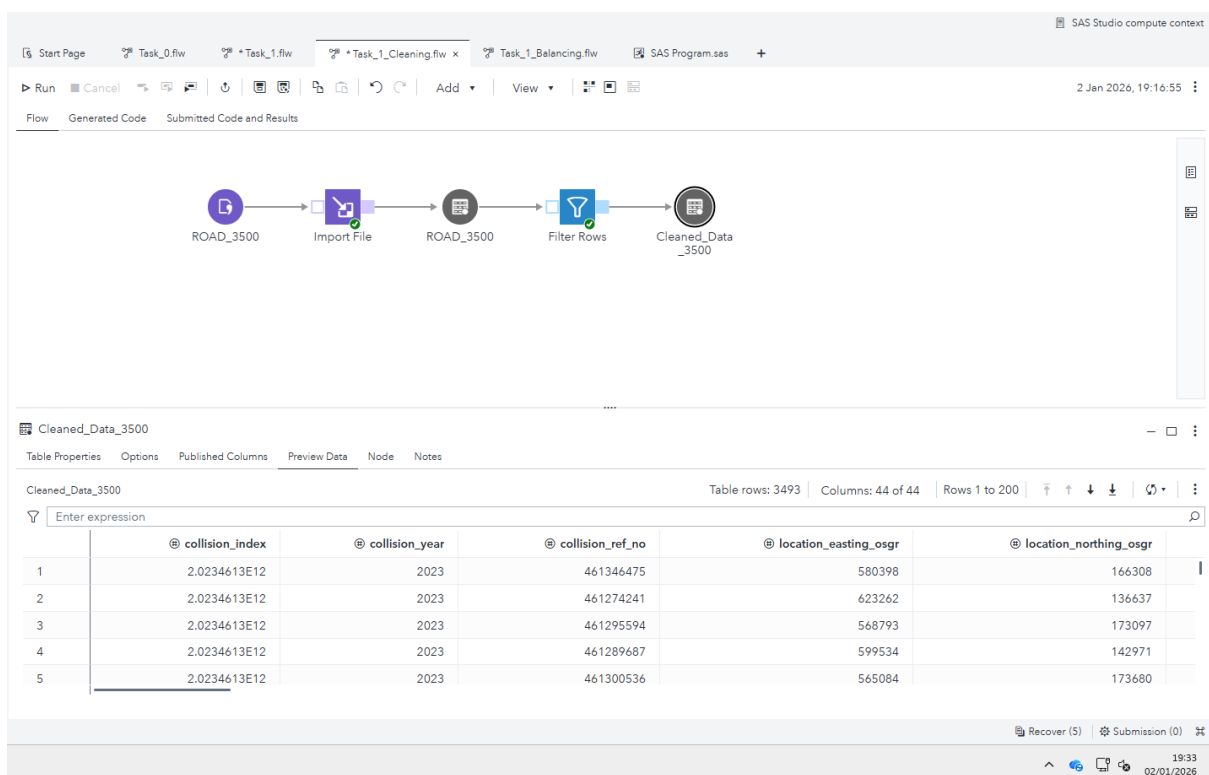
Exploratory analysis and feature association outputs for local authority highway classification, highlighting geographic and road-related factors influencing accident distribution.

B5: Collision Severity Analysis



Analysis of collision severity, including factor importance and relationships between severity levels and the number of casualties across local authority highways.

B6: Data Cleaning Workflow



SAS workflow and validation outputs illustrating data cleaning steps, including removal of invalid records and confirmation of the cleaned dataset structure. Cleaned dataset preview (3493 rows)

B7: Date & Month Feature Engineering

SAS Studio compute context

Start Page Task_0.flw Task_1.flw Task_1_Cleaning.flw Task_1_Balancing.flw Import.cti SAS Program.sas x +

Run Cancel Copy to My Snippets Code to Flow Debug Clear Log Jan 4, 2026, 11:49:23 AM

Code

```

1 data work.recodedRanges0002;
2 length _months_ $3;
3 set WORK.IMPORT;
4
5 /* Convert the date variable to SAS date value */
6 sas_date = mdy(month(date), day(date), year(date));
7
8 /* Check month ranges and assign labels accordingly */
9 if ('01JAN2023'd <= sas_date <= '31JAN2023'd) then _months_ = 'jan';
10 else if ('01FEB2023'd <= sas_date <= '28FEB2023'd) then _months_ = 'feb';
11 else if ('01MAR2023'd <= sas_date <= '31MAR2023'd) then _months_ = 'mar';
12 else if ('01APR2023'd <= sas_date <= '30APR2023'd) then _months_ = 'apr';
13 else if ('01MAY2023'd <= sas_date <= '31MAY2023'd) then _months_ = 'may';
14 else if ('01JUN2023'd <= sas_date <= '30JUN2023'd) then _months_ = 'jun';
15 else if ('01JUL2023'd <= sas_date <= '31JUL2023'd) then _months_ = 'jul';
16 else if ('01AUG2023'd <= sas_date <= '31AUG2023'd) then _months_ = 'aug';
17 else if ('01SEP2023'd <= sas_date <= '30SEP2023'd) then _months_ = 'sep';
18 else if ('01OCT2023'd <= sas_date <= '31OCT2023'd) then _months_ = 'oct';
19 else if ('01NOV2023'd <= sas_date <= '30NOV2023'd) then _months_ = 'nov';
20 else if ('01DEC2023'd <= sas_date <= '31DEC2023'd) then _months_ = 'dec';
21 else _months_ = 'other';
22
23 format sas_date date9.; /* Optional: Set the format for display purposes */
24 run;
25

```

Log Output Data (1)

RECODEDRANGES0002 Table rows: 3493 Columns: 46 of 46 Rows 1 to 200

	@ coll_adju_sev_slig	@ sas_date
1	1	14JUL2023
2	1	05FEB2023
3	0	17MAR2023
4	1	19MAR2023
5	1	17APR2023
6	1	22APR2023
7	1	22MAY2023
8	1	03JUL2023
9	1	30JUL2023
10	0	29AUG2023
11	1	17SEP2023
12	1	20OCT2023
13	1	23DEC2023
14	1	10JUL2023
15	0	12JUL2023
16	1	25SEP2023
17	1	01MAR2023

SAS Studio compute context

Start Page Task_0.flw Task_1.flw Task_1_Cleaning.flw Task_1_Balancing.flw Import.cti SAS Program.sas x +

Run Cancel Copy to My Snippets Code to Flow Debug Clear Log Jan 4, 2026, 11:49:23 AM

Code

```

1 data work.recodedRanges0002;
2 length _months_ $3;
3 set WORK.IMPORT;
4
5 /* Convert the date variable to SAS date value */
6 sas_date = mdy(month(date), day(date), year(date));
7
8 /* Check month ranges and assign labels accordingly */
9 if ('01JAN2023'd <= sas_date <= '31JAN2023'd) then _months_ = 'jan';
10 else if ('01FEB2023'd <= sas_date <= '28FEB2023'd) then _months_ = 'feb';
11 else if ('01MAR2023'd <= sas_date <= '31MAR2023'd) then _months_ = 'mar';
12 else if ('01APR2023'd <= sas_date <= '30APR2023'd) then _months_ = 'apr';
13 else if ('01MAY2023'd <= sas_date <= '31MAY2023'd) then _months_ = 'may';
14 else if ('01JUN2023'd <= sas_date <= '30JUN2023'd) then _months_ = 'jun';
15 else if ('01JUL2023'd <= sas_date <= '31JUL2023'd) then _months_ = 'jul';
16 else if ('01AUG2023'd <= sas_date <= '31AUG2023'd) then _months_ = 'aug';
17 else if ('01SEP2023'd <= sas_date <= '30SEP2023'd) then _months_ = 'sep';
18 else if ('01OCT2023'd <= sas_date <= '31OCT2023'd) then _months_ = 'oct';
19 else if ('01NOV2023'd <= sas_date <= '30NOV2023'd) then _months_ = 'nov';
20 else if ('01DEC2023'd <= sas_date <= '31DEC2023'd) then _months_ = 'dec';
21 else _months_ = 'other';
22
23 format sas_date date9.; /* Optional: Set the format for display purposes */
24 run;
25

```

Log Output Data (1)

RECODEDRANGES0002 Table rows: 3493 Columns: 46 of 46 Rows 1 to 200

	@ _months_	@ collision_index	@ collision_year
1	jul	2.0234613E12	2023
2	feb	2.0234613E12	2023
3	mar	2.0234613E12	2023
4	mar	2.0234613E12	2023
5	apr	2.0234613E12	2023
6	apr	2.0234613E12	2023
7	may	2.0234613E12	2023
8	jul	2.0234613E12	2023
9	jul	2.0234613E12	2023
10	aug	2.0234613E12	2023
11	sep	2.0234614E12	2023
12	oct	2.0234614E12	2023
13	dec	2.0234614E12	2023
14	jul	2.0234613E12	2023
15	jul	2.0234613E12	2023
16	sep	2.0234614E12	2023
17	mar	2.0234613E12	2023

SAS code, outputs, and validation results for the transformation of collision date variables and the creation of calendar month features to support temporal analysis.

Files

- code_date_month.pdf
- Date_month_coversion.png
- Date_month_2.png

- results_date_month.pdf
- summary_date_month.pdf
- log_date_month.pdf

B8: Time Feature Engineering

The screenshot displays the SAS Studio environment. The left pane shows the following SAS code:

```

1 data work.recodedRanges0002;
2   length time $;
3   format time time8.;
4   set WORK.IMPORT1;
5   /* Extract the hour from the time variable */
6   hour_of_day = hour(time);
7   /* Categorize time into different periods */
8   if 0 <= hour_of_day < 6 then time_category = 'night';
9   else if 6 <= hour_of_day < 12 then time_category = 'morning';
10  else if 12 <= hour_of_day < 18 then time_category = 'afternoon';
11  else if 18 <= hour_of_day <= 23 then time_category = 'evening';
12  else time_category = 'other';
13  run;

```

The right pane shows the output table 'RECODEDRANGES0002' with 3493 rows and 48 columns. The first 17 rows are displayed:

	sas_date	hour_of_day	time_category
1	14JUL2023	20	eveni
2	05FEB2023	10	morni
3	17MAR2023	18	eveni
4	19MAR2023	10	morni
5	17APR2023	9	morni
6	22APR2023	17	after
7	22MAY2023	17	after
8	03JUL2023	5	night
9	30JUL2023	14	after
10	29AUG2023	14	after
11	17SEP2023	17	after
12	20OCT2023	18	eveni
13	23DEC2023	0	night
14	10JUL2023	17	after
15	12JUL2023	8	morni
16	25SEP2023	16	after
17	01MAR2023	21	eveni

Extraction and categorisation of collision time variables, including hour-of-day and time-of-day features, with supporting SAS outputs and execution logs.

Files

- code_sas_time.pdf
- listings_sas_time.pdf
- results_sas_time.pdf
- summary_sas_time.pdf
- SAS_time_3.png
- log_sas_time.pdf

B9: Data Balancing

The screenshot displays the SAS Studio interface. The left pane shows the SAS code editor with the following code:

```

6 run;
7
8 proc freq data=WORK.IMPORT2;
9   tables collision_severity;
10 run;
11
12 proc surveyselect data=sev3
13   out=sev3_sampled
14   method=srs
15   sampsize=705
16   seed=6859062;
17 run;
18
19 data Road_3500_Balanced;
20   set sev1 sev2 sev3_sampled;
21 run;
22
23 proc surveyselect data=Road_3500_Balanced
24   out=Road_3500_Balanced_Final
25   method=srs
26   samprate=1
27   seed=6859062;
28 run;
29
30 proc freq data=Road_3500_Balanced_Final;
31   tables collision_severity;
32 run;
33
34 proc contents data=WORK.Road_3500_Balanced_Final;
35 run;
36
37 proc export data=WORK.Road_3500_Balanced_Final
38   outfile="/export/viya/homes/st01870@surrey.ac.uk/casuser/Road_3500_Balanced_
39   dbms=csv
40   replace;
41 run;
42
43

```

The right pane shows the 'Log' tab with the output of the PROC CONTENTS procedure:

The CONTENTS Procedure

Data Set Name	WORK.ROAD_3500_BALANCED_FINAL	Observations	1451
Member Type	DATA	Variables	48
Engine	V9	Indexes	0
Created	01/04/2026 22:43:14	Observation Length	384
Last Modified	01/04/2026 22:43:14	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64, LINUX_POWER_64		
Encoding	utf-8 Unicode (UTF-8)		

Below the PROC CONTENTS output, the 'Engine/Host Dependent Information' is shown:

Engine/Host Dependent Information	
Data Set Page Size	65536
Number of Data Set Pages	9
First Data Page	1
Max Obs per Page	170
Obs in First Data Page	148
Number of Data Set	0

Data balancing procedure applied to collision severity classes, including sampling logic, post-balancing validation, and confirmation of the final balanced dataset.

Files

- code_task1_databalancing.pdf
- listing_task1_databalancing.pdf
- results_task1_databalancing.pdf
- summary_task1_databalancing.pdf
- log_task1_databalancing.pdf
- Data_Balancing.png

B10: SAS Logs

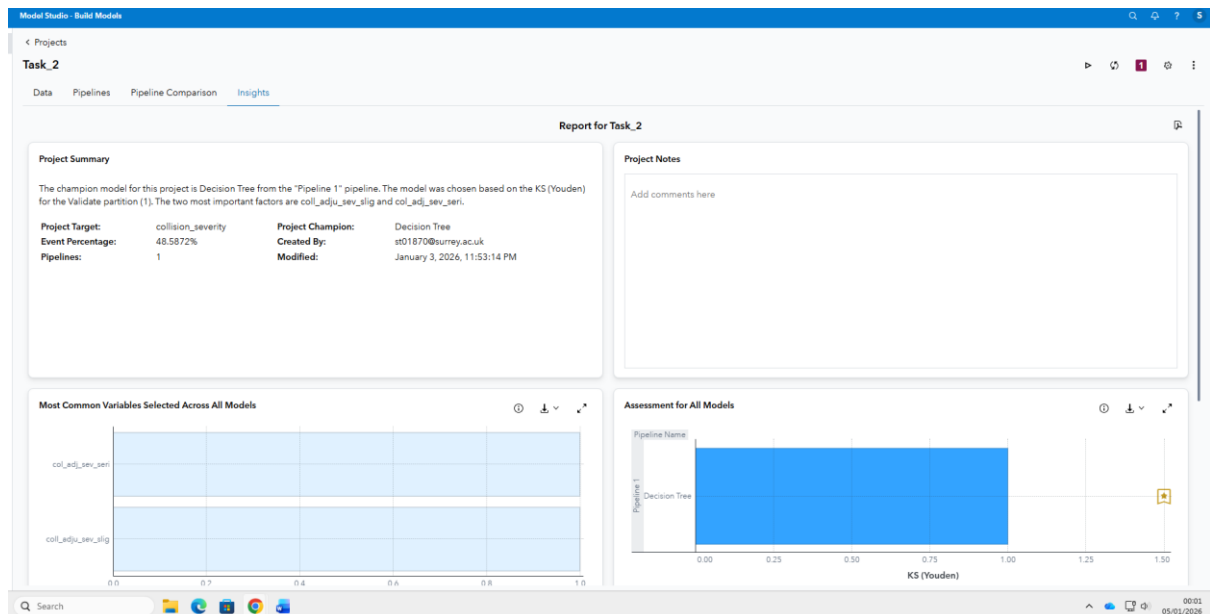
SAS execution logs providing technical evidence of successful completion of Task 1 processes and reproducibility of results.

Files

- Task1_log_file.pdf

APPENDIX - C: Task 2 - Predictive Modelling

C1: Modelling Dataset and Variable Roles



Project Settings

Partition Data

☒ Create partition variable

Note: These settings are active only when a partition variable is not set within the data. Using a data source with a pre-defined partition variable or manually selecting a partition variable will override these settings.

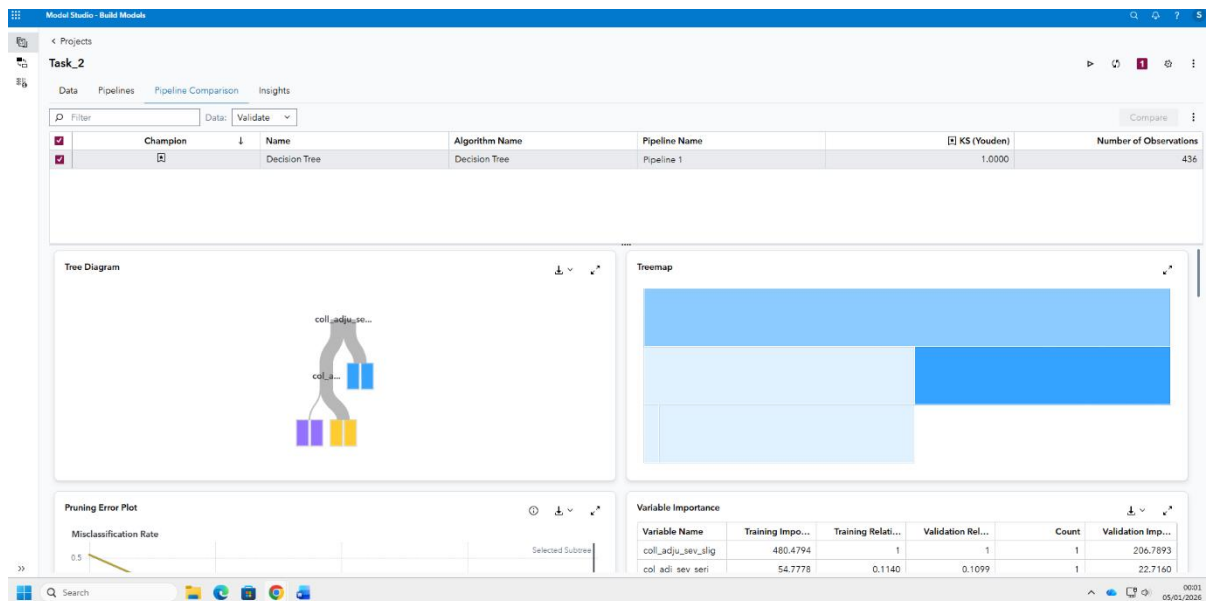
Method: **Stratify**

Training: **70** 70.00%

Validation: **30** 30.00%

Test: **0** 0.00%

Save **Cancel**



Modelling dataset configuration showing the balanced dataset, variable role assignments, and stratified training-validation partition settings used for predictive modelling.

- Screenshot 2026-01-05 000130.png
(Dataset overview showing ROAD_3500_BALANCED_FINAL with rows and variables)
- task2_2.png
(Variable role assignments: Target, Input, Rejected)
- Screenshot 2026-01-05 000112.png
(Partition settings showing stratified 70% training and 30% validation)

C2: Decision Tree Outputs

Decision Tree model outputs, including tree structure, variable importance, and validation performance used to identify key predictors of collision severity.

- Decision_Tree.pdf
(Include pages showing tree diagram, variable importance, pruning error plot, and validation performance)

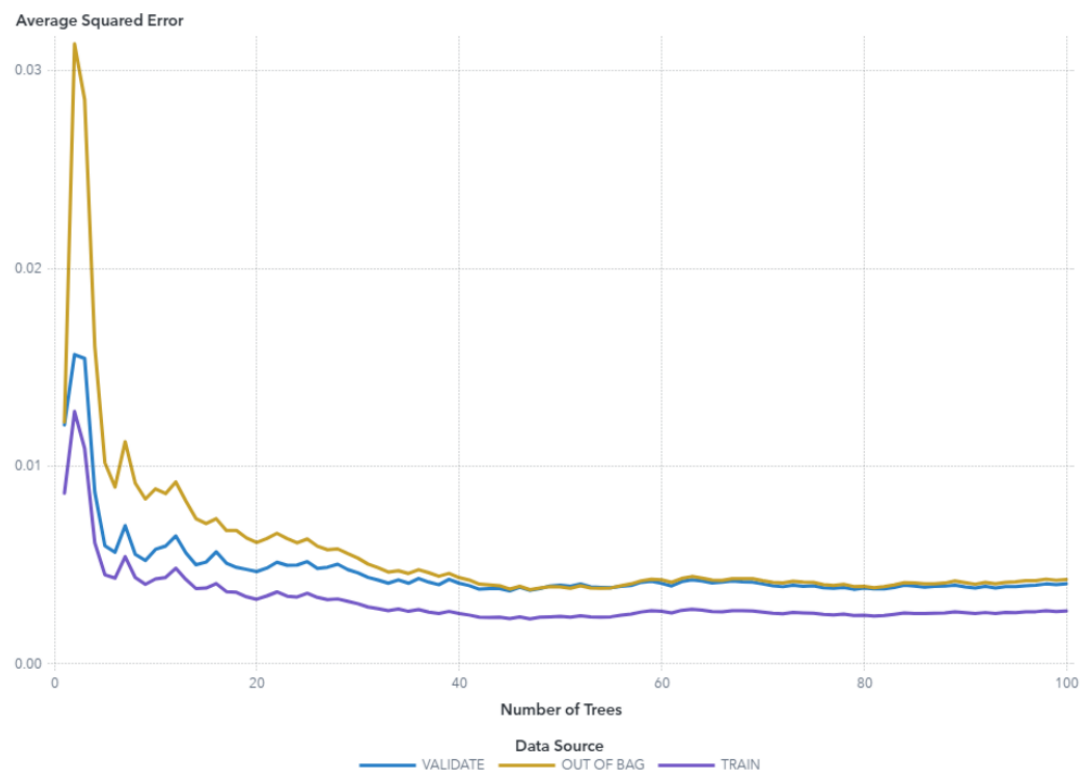
C3: Logistic Regression Outputs

Multinomial Logistic Regression model outputs, including classification performance and accuracy statistics used for comparison with tree-based models.

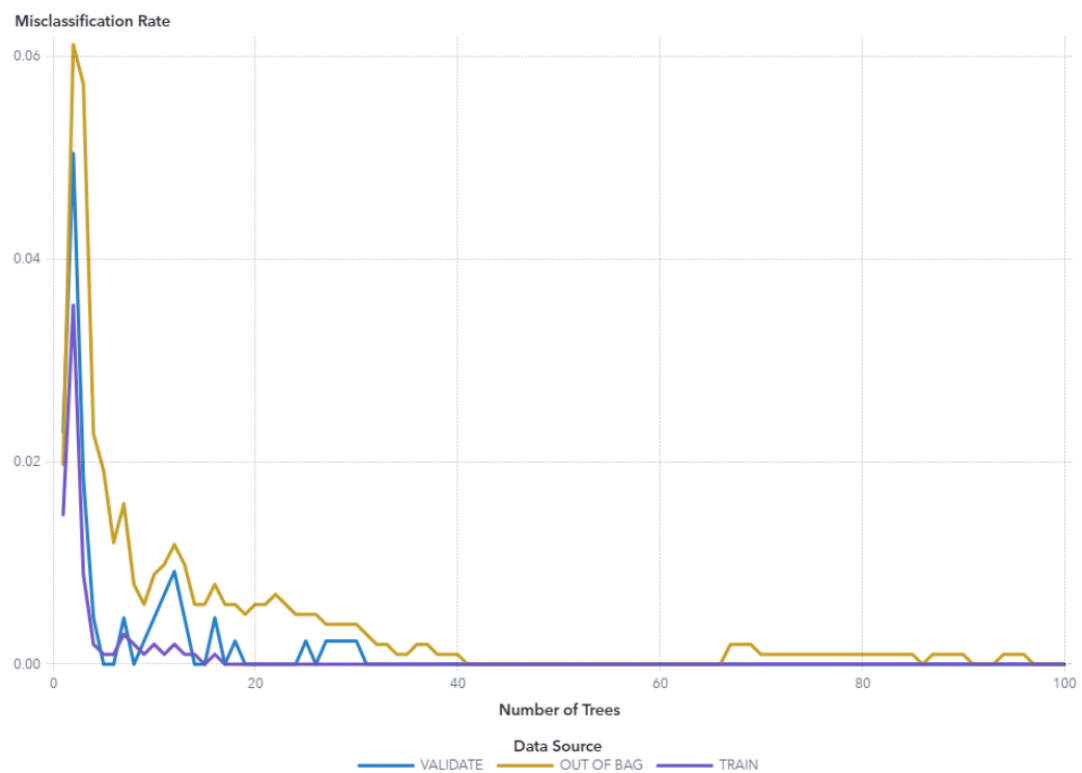
- Logistic_Regression.pdf
(Include only the pages showing classification table, accuracy/misclassification statistics, and KS performance. Exclude full coefficient tables unless summarised elsewhere.)

C4: Random Forest Outputs

Average Squared Error



Misclassification Rate

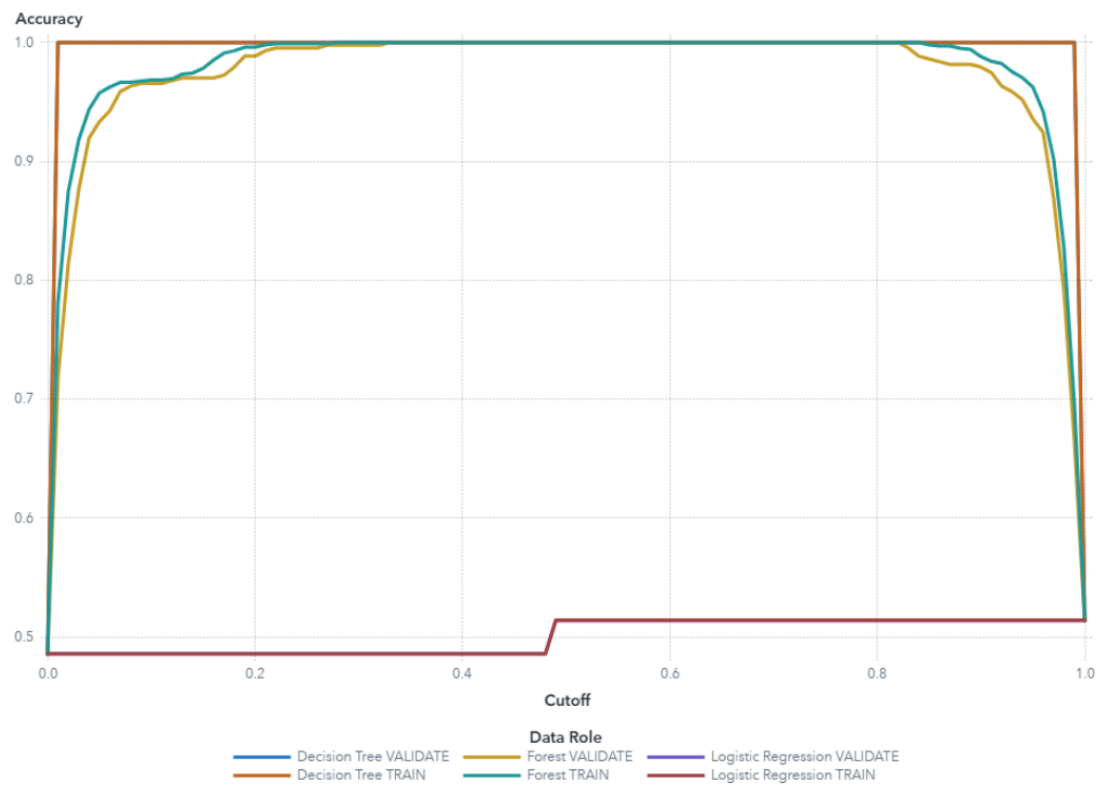


Random Forest model outputs summarising Average Squared Error and Misclassification Rate for comparison with other classification models.

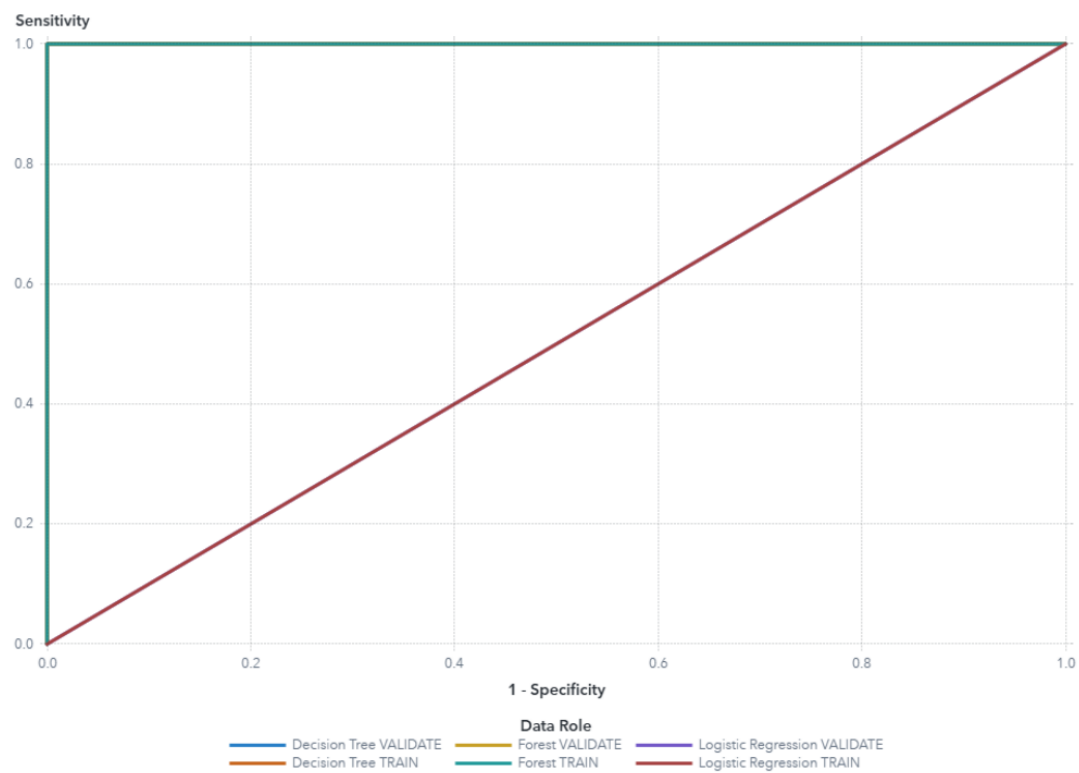
- Forest.pdf
(Include pages showing model performance metrics and variable importance summaries)

C5: Model Comparison Results

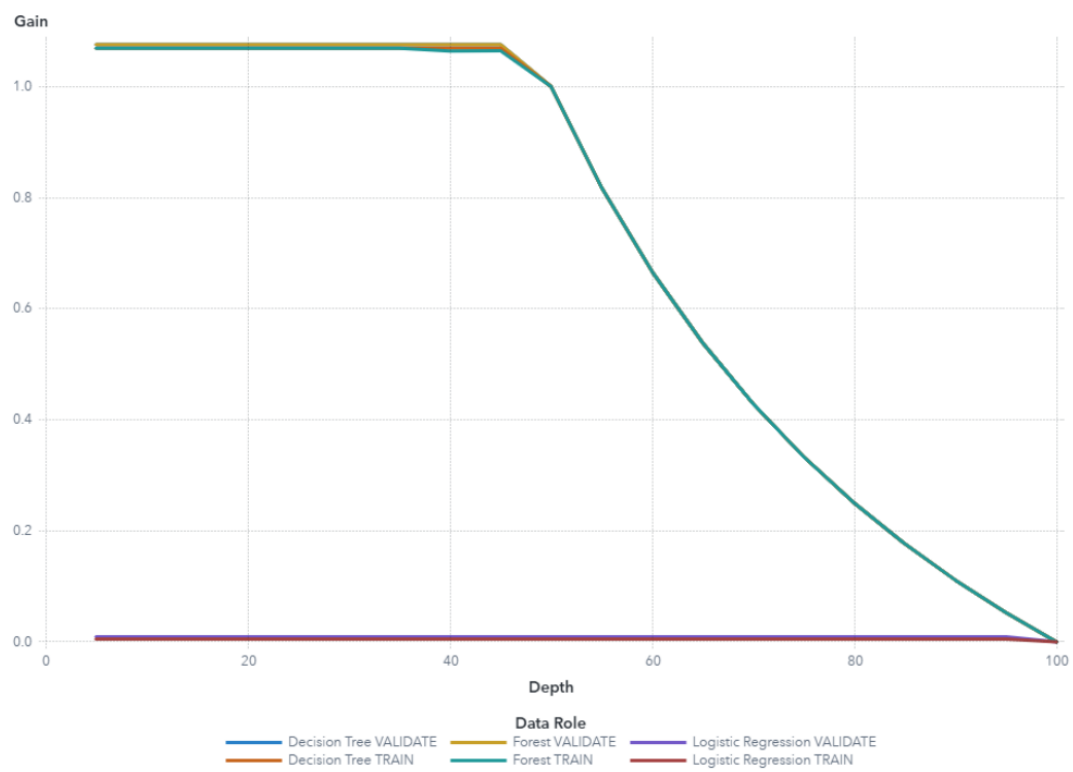
Accuracy



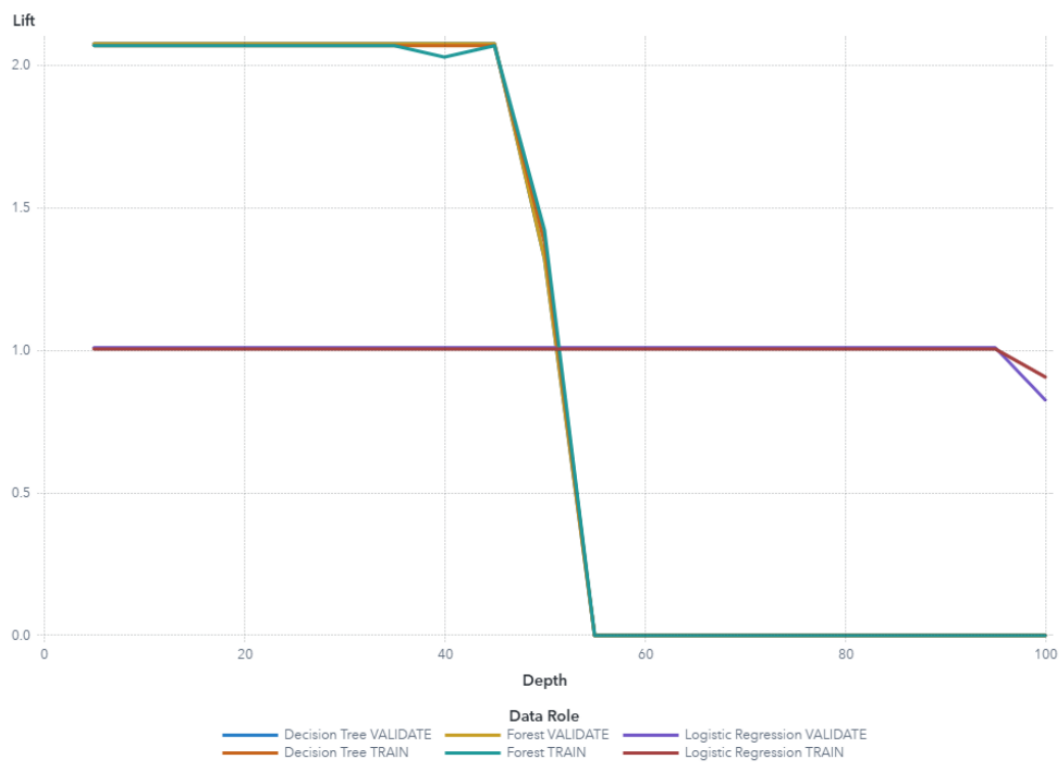
ROC



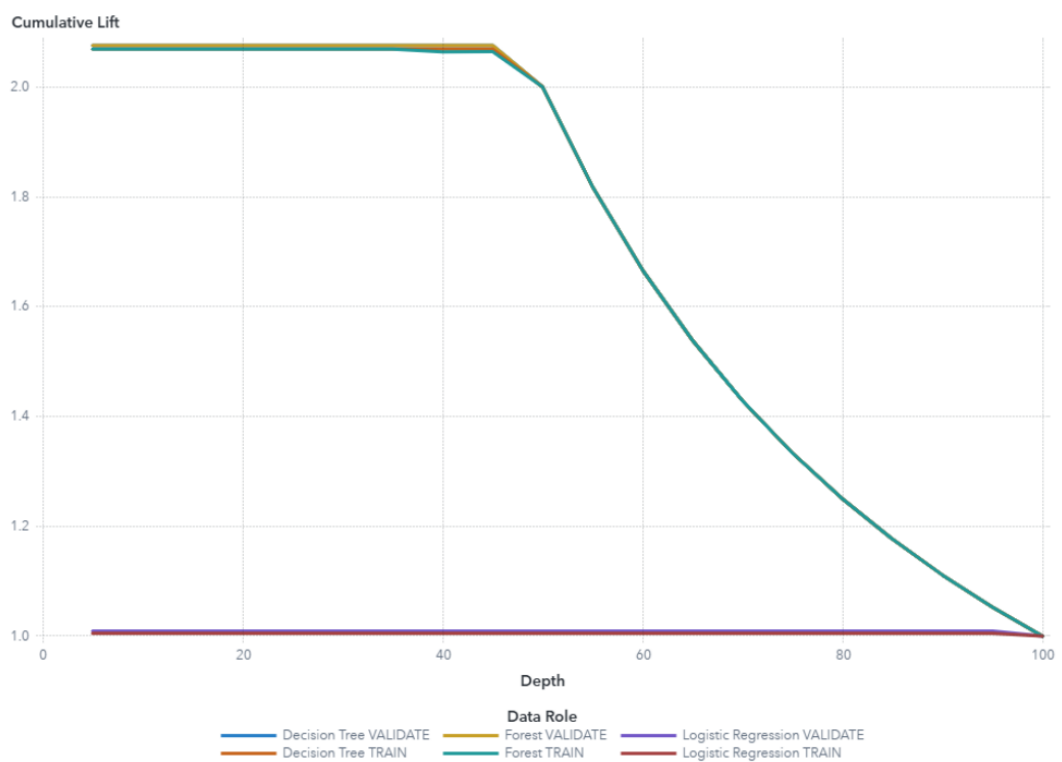
Gain



Lift



Cumulative Lift



Model Comparison

Champion	Name	Algorithm Name	KS (Youden)	Accuracy
true	Decision Tree	Decision Tree	1	1
false	Logistic Regression	Logistic Regression	0	0.5138
false	Forest	Forest	1	1

Average Squared Error	Area Under ROC	Cumulative Lift	Cumulative Captured Response Percentage	Cutoff
0	1	2.0755	20.7547	0.5000
0.1755	0.5000	1.0092	10.0917	0.5000
0.0041	1	2.0755	20.7547	0.5000

Data Role	Depth	F1 Score	False Discovery Rate	False Positive Rate
VALIDATE	10	1	0	0
VALIDATE	10	0		0
VALIDATE	10	1	0	0

Gain	Gini Coefficient	ROC Separation	Lift	Misclassification Rate
1.0755	1	1	2.0755	0
0.0092	0	0	1.0092	0.5138

Gain	Gini Coefficient	ROC Separation	Lift	Misclassification Rate
1.0755	1	1	2.0755	0

Multi-Class Log Loss	Misclassification at Cutoff	Misclassification Rate (Event)	Number of Observations	Root Average Squared Error
0.0000	0	0	436	0
0.8001	0.5138	0.4862	436	0.4189
0.0476	0	0	436	0.0637

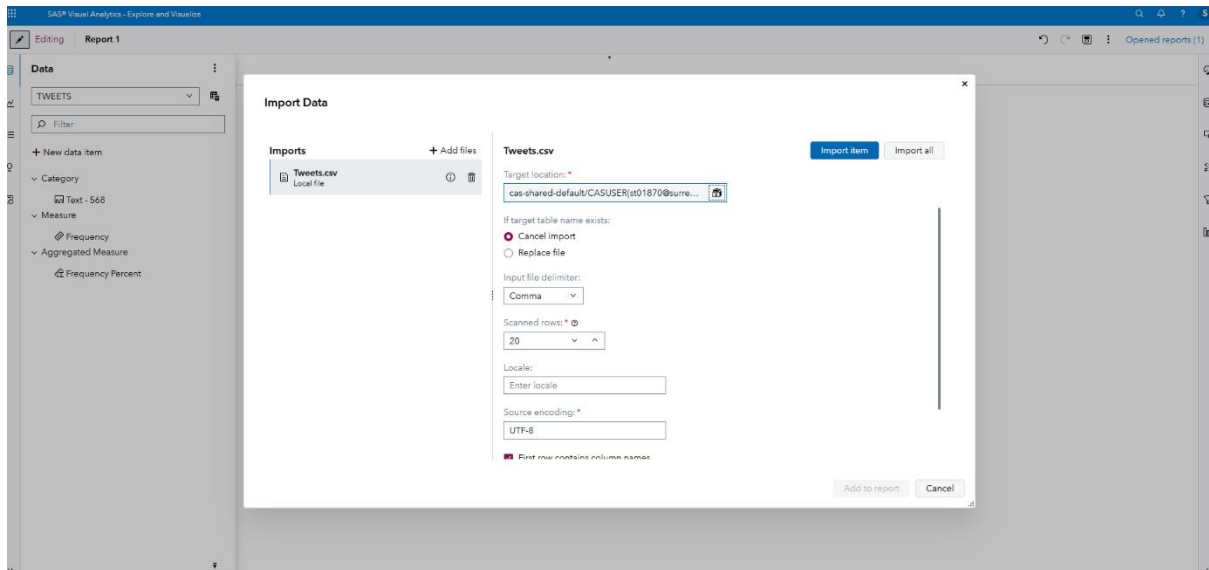
Captured Response Percentage
10.3774
5.0459
10.3774

Model comparison results across Logistic Regression, Decision Tree, and Random Forest models, including KS (Youden) statistics and evidence supporting champion model selection.

- Model_Comparison.pdf
(Include page showing KS (Youden) statistic, champion model selection, and validation performance)
- Insights.pdf
(Include page showing project summary and assessment for all models)

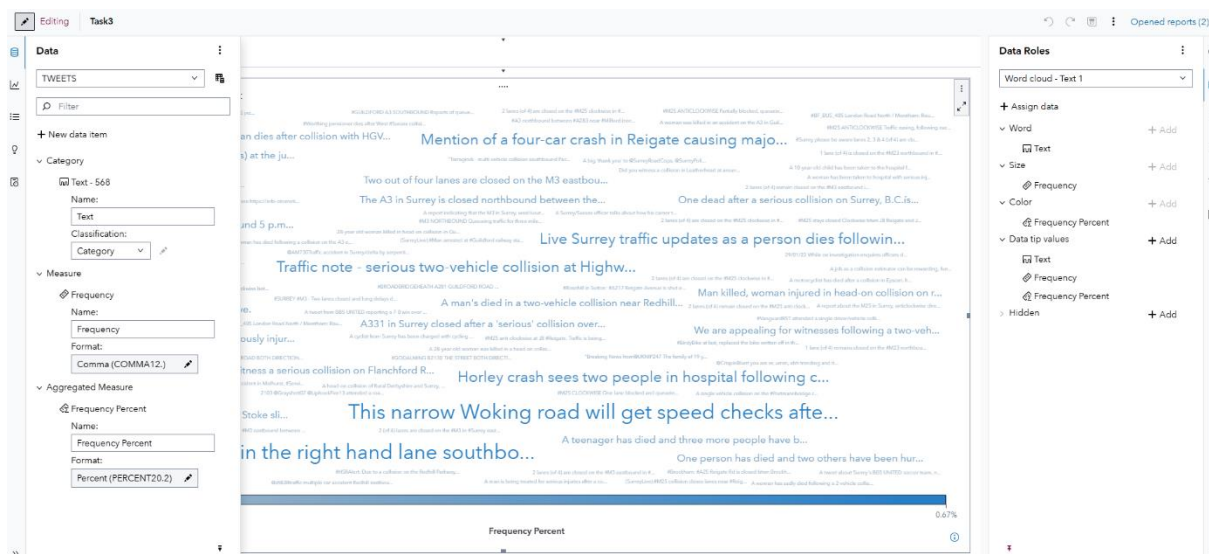
APPENDIX - D : Task 3 - Text Analytics

D1: Data Import and Variable Roles



Import of the Twitter dataset into SAS Visual Analytics, showing successful data loading and assignment of the tweet text variable with the appropriate Text role for downstream text analytics processing.

D2: Word Cloud Visualisation



Word cloud visualisation illustrating the most frequent terms appearing in traffic-related tweets, highlighting dominant themes such as collisions, road closures, and traffic disruptions.

D3: Text Analytics Pipeline

SAS Visual Text Analytics pipeline illustrating the sequential processing of tweet data, including concept extraction, text parsing, sentiment analysis, topic modelling, and category classification.

- File: Concepts.pdf

D4: Concept Extraction Outputs

Outputs from the concept extraction node, identifying key entities and patterns related to road traffic incidents, locations, and event descriptors within the tweet corpus.

- File: Concepts.pdf

D5: Text Parsing Outputs

Results of text parsing, including tokenisation, term frequency generation, and linguistic processing applied to raw tweet text prior to sentiment and topic analysis.

- File: Text_Parsing.pdf

D6: Sentiment Analysis Outputs

Sentiment analysis outputs generated using the default English sentiment model, classifying tweets into positive, neutral, and negative sentiment categories.

- File: Sentiment.pdf

D7: Topic Modelling Outputs

Topic modelling results identifying latent themes within traffic-related tweets, supporting the interpretation of common incident types and public response patterns.

- File: Topics.pdf

D8: Category Classification Outputs

Category classification outputs grouping tweets into predefined thematic categories, enabling structured interpretation of traffic incidents and related discussions.

- File: Categories.pdf